

ECE2150J Intro to Circuits

Chapter 12. Three-Phase Circuits

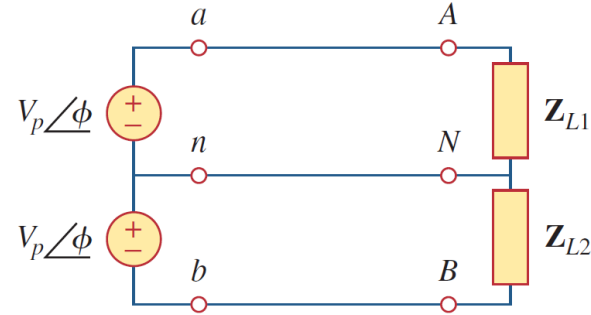
*Recommend you to bring your calculator to lectures

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12.1 Introduction



Single phase two wire

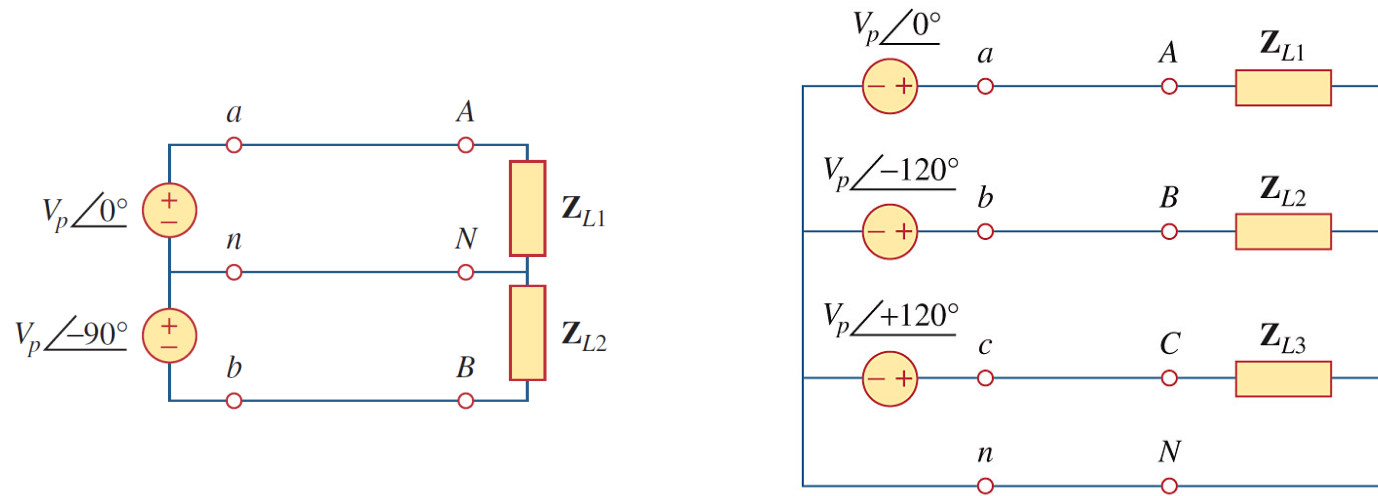


Single phase three wire

A single-phase ac power system: a generator connected through a pair of wires to a load.

A single-phase three wire system: two identical sources (equal magnitude and the same phase) connected to two loads by two outer wires and the neutral.

Polyphase: AC sources operate at the same frequency but different phases.



Two-phase three-wire system Three-phase four-wire system

A three-phase system: Three sources having the same amplitude and frequency but out of phase with each other by 120° . The three-phase system is **by far the most prevalent and most economical polyphase system.**

Three-phase systems are important for at least three reasons.

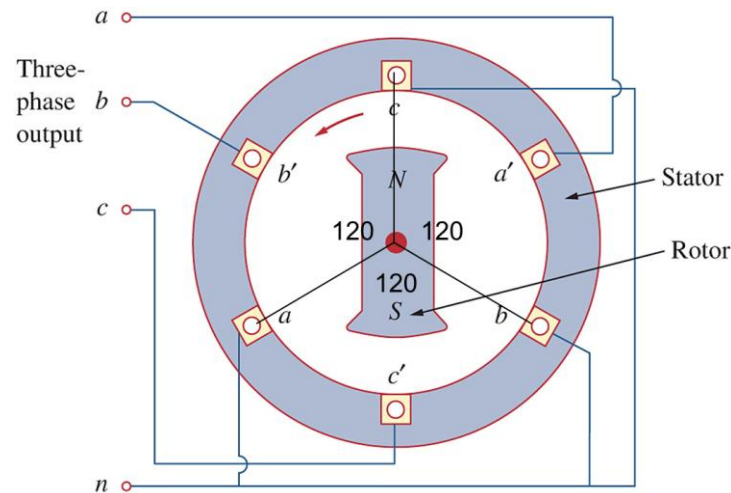
First, nearly all electric power is generated and distributed in three-phase, at the operating frequency of 50 or 60 Hz. **Multiphases** can be generated: from 1-2 to even 48 phases.

Second, the instantaneous power in a three-phase system can be **constant** (not pulsating). This results in uniform power transmission and less vibration of three-phase machines.

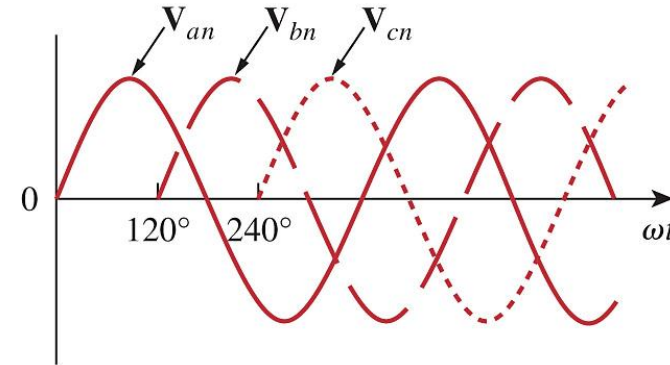
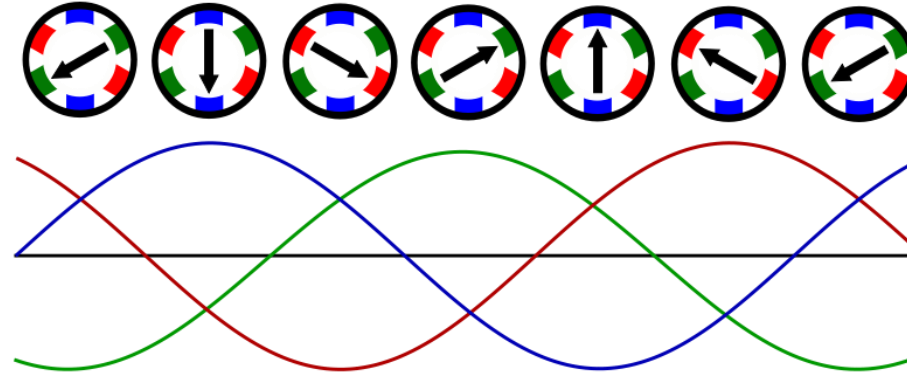
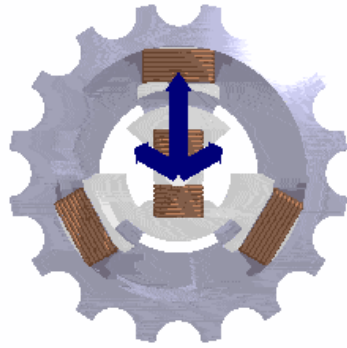
Third, for the same amount of power, the three-phase system is **more economical** than the single phase.

12.2 Balanced Three-Phase Voltages

Three-phase voltages are often produced with a three-phase ac generator.



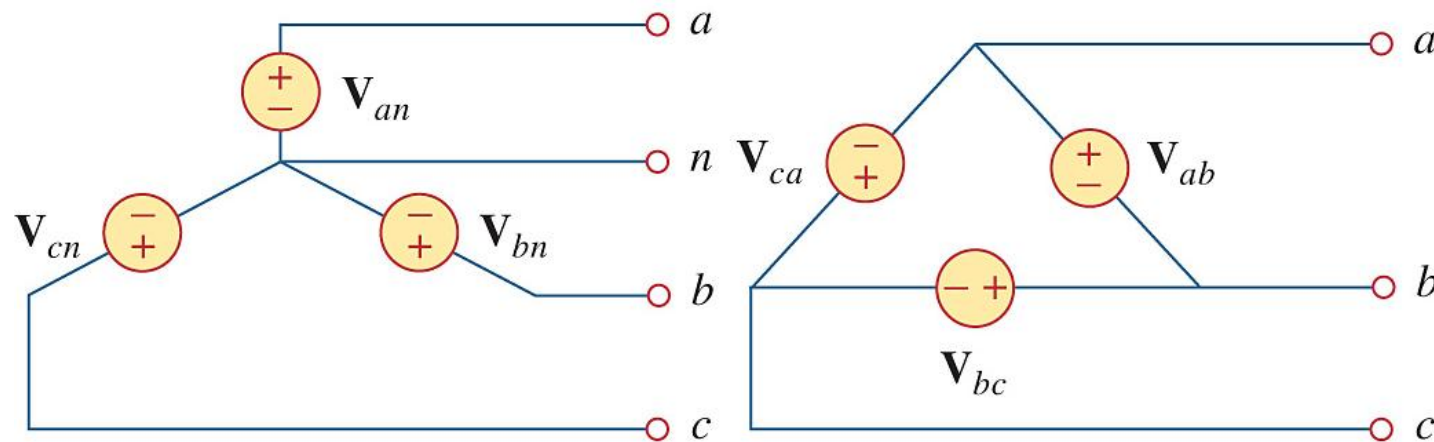
Rotor: a rotating magnet
Stator: a stationary winding

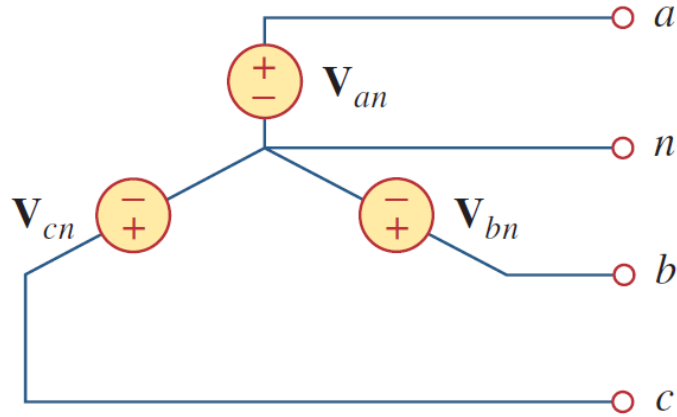


As the rotator rotates, the flux through each coil varies sinusoidally with time, inducing a sinusoidal voltage. Because the coils are placed **120°** apart, the induced voltages are out of phase by **120°**.

Voltage sources

A typical three-phase system consists of three voltage sources connected to loads by three or four wires. A three-phase system is equivalent to three single-phase system. **The voltage sources** can be either **wye-** or **delta-connected**.





$\mathbf{V}_{an}$, $\mathbf{V}_{bn}$, and $\mathbf{V}_{cn}$ are phase voltages.

Balanced phase voltages: Equal in magnitude and frequency and are out of phase with each other by 120° . This implies that

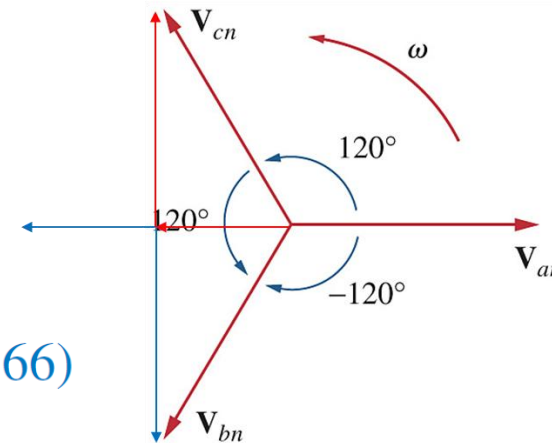
(i) $|\mathbf{V}_{an}| = |\mathbf{V}_{bn}| = |\mathbf{V}_{cn}|$

(ii) $\mathbf{V}_{an} + \mathbf{V}_{bn} + \mathbf{V}_{cn} = 0$

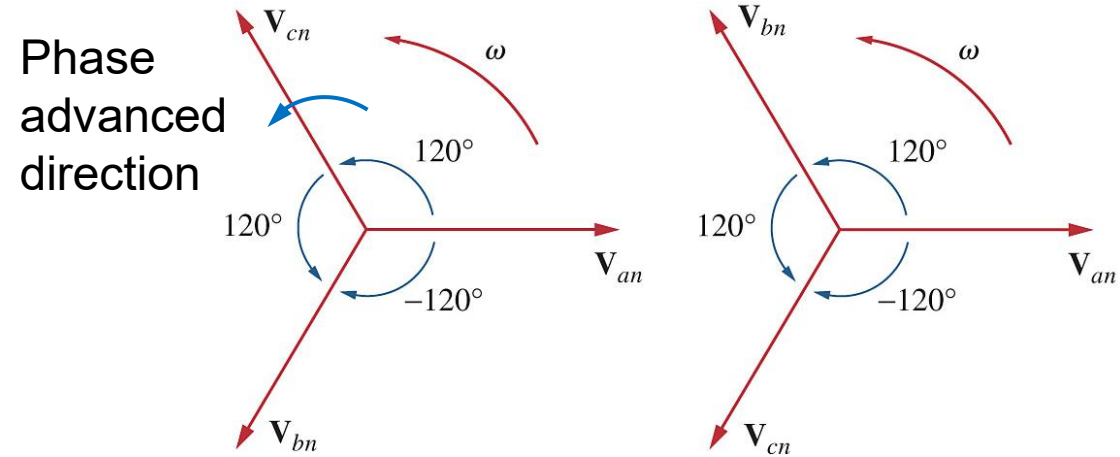
$$= V_p \angle 0^\circ + V_p \angle -120^\circ + V_p \angle +120^\circ$$

$$= V_p(1.0 - 0.5 - j0.866 - 0.5 + j0.866)$$

$$= 0$$



Phase sequence



$$V_{an} = V_p \angle 0^\circ$$

$$V_{bn} = V_p \angle -120^\circ$$

$$V_{cn} = V_p \angle -240^\circ = V_p \angle +120^\circ$$

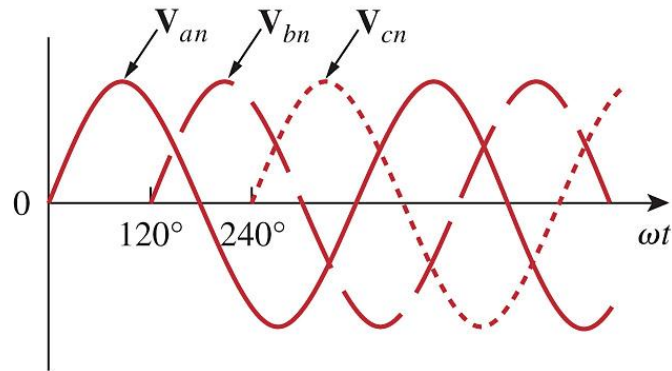
$$V_{an} = V_p \angle 0^\circ$$

$$V_{cn} = V_p \angle -120^\circ$$

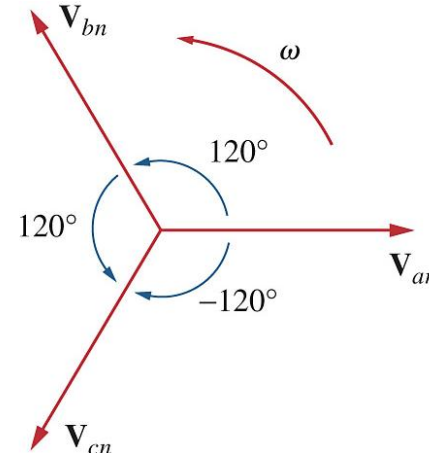
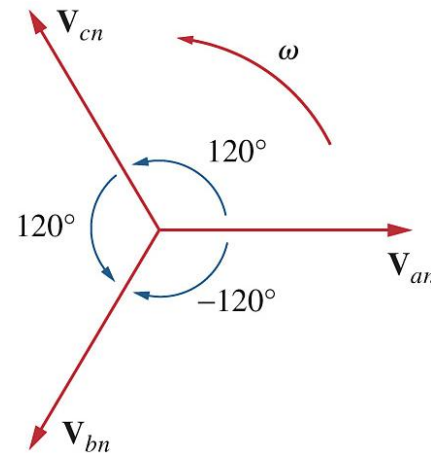
$$V_{bn} = V_p \angle -240^\circ = V_p \angle +120^\circ$$

*** V_p is the effective (rms) value of the phase voltage**

The phase sequence is the time order in which the voltages pass through their respective maximum values.

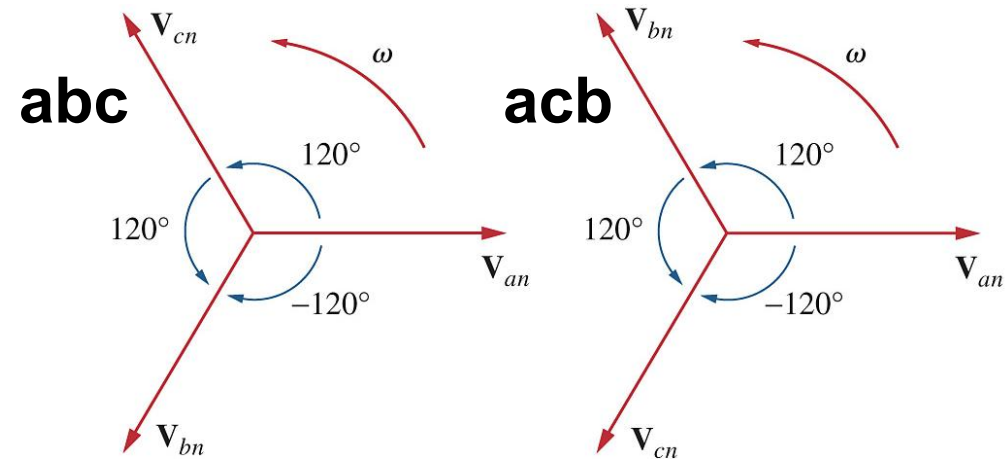


abc: a leads b, b leads c
(i.e., $\angle a > \angle b > \angle c$)



From which phase the voltage curve above is generated?
abc or acb sequence?

The phase sequence is determined **by the order in which the phasors pass through a fixed point** in the phase diagram.



$$\mathbf{V}_{an} = V_p \angle 0^\circ$$

$$\mathbf{V}_{bn} = V_p \angle -120^\circ$$

$$\mathbf{V}_{cn} = V_p \angle -240^\circ = V_p \angle +120^\circ$$

$$\mathbf{V}_{an} = V_p \angle 0^\circ$$

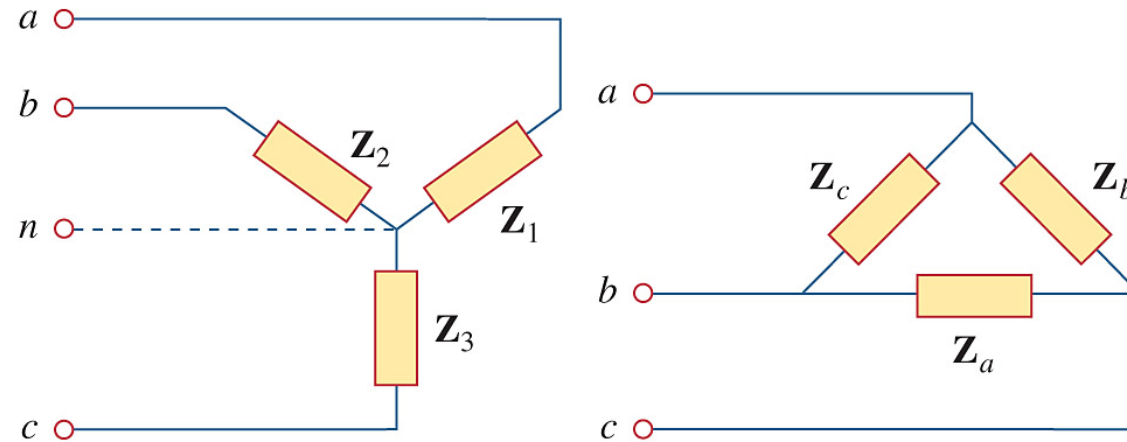
$$\mathbf{V}_{cn} = V_p \angle -120^\circ$$

$$\mathbf{V}_{bn} = V_p \angle -240^\circ = V_p \angle +120^\circ$$

* V_p is the effective (rms) value of the phase voltage

Phase load connection

Similar to the source connections, a three-phase load can be either wye-connected or delta-connected



Balanced Load

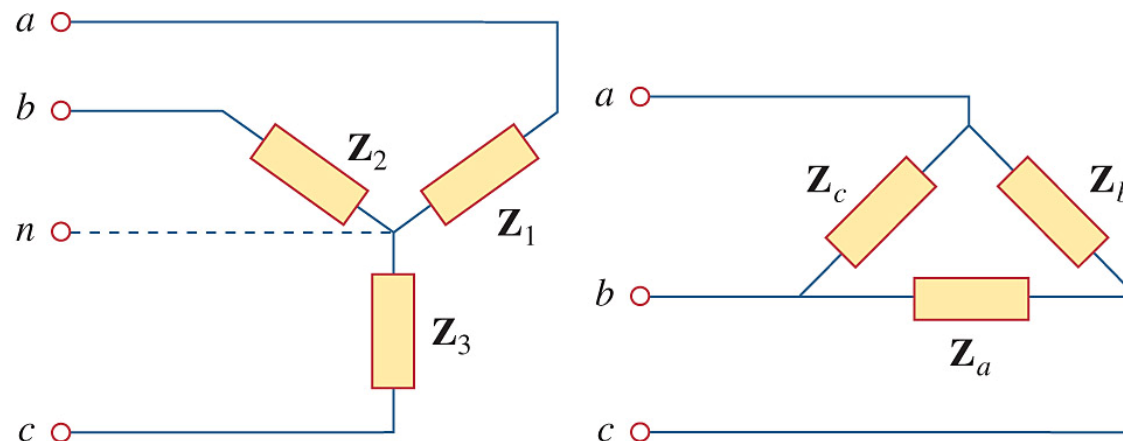
A balanced load is one in which the phase impedances are **equal in magnitude and in phase**.

(i) Balanced wye-connected load:

$$\mathbf{Z}_1 = \mathbf{Z}_2 = \mathbf{Z}_3 = \mathbf{Z}_Y$$

(i) Balanced delta-connected load:

$$\mathbf{Z}_a = \mathbf{Z}_b = \mathbf{Z}_c = \mathbf{Z}_\Delta$$



A wye-connected load can be transformed into a delta-connected load, or vice versa.

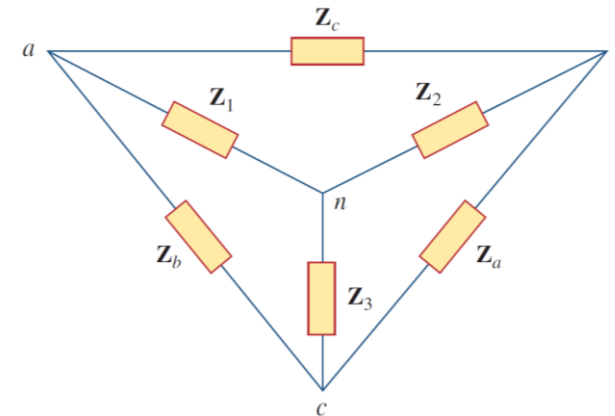
$$Z_{\Delta} = 3Z_Y \text{ or } Z_Y = \frac{1}{3} Z_{\Delta}$$

Y- Δ Conversion:

$$\begin{aligned} Z_a &= \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_1} \\ Z_b &= \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_2} \\ Z_c &= \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_3} \end{aligned}$$

Δ -Y Conversion:

$$\begin{aligned} Z_1 &= \frac{Z_b Z_c}{Z_a + Z_b + Z_c} \\ Z_2 &= \frac{Z_c Z_a}{Z_a + Z_b + Z_c} \\ Z_3 &= \frac{Z_a Z_b}{Z_a + Z_b + Z_c} \end{aligned}$$



$$Z_1 = Z_2 = Z_3 = Z_Y$$

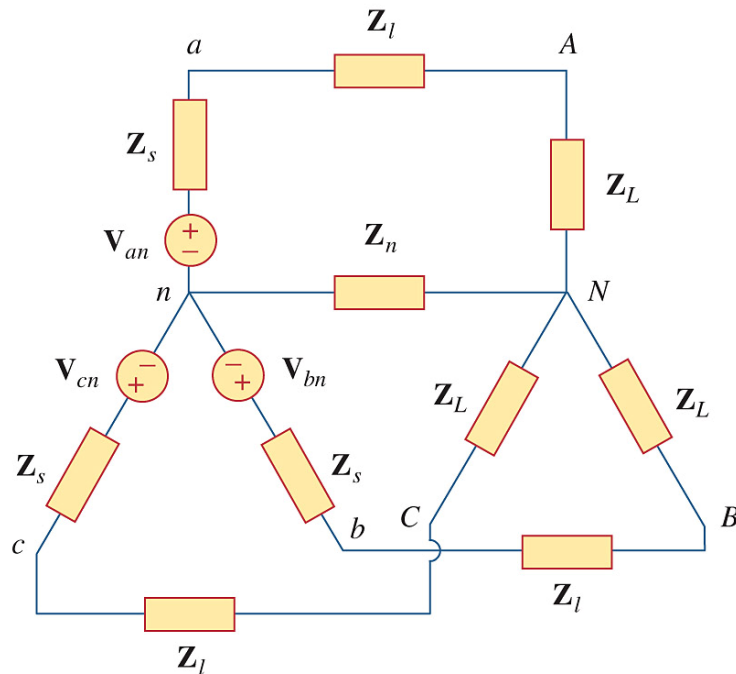
$$Z_a = Z_b = Z_c = Z_{\Delta}$$

Since both the three-phase source and the three-phase load can be either wye- or delta-connected, we have four possible connections:

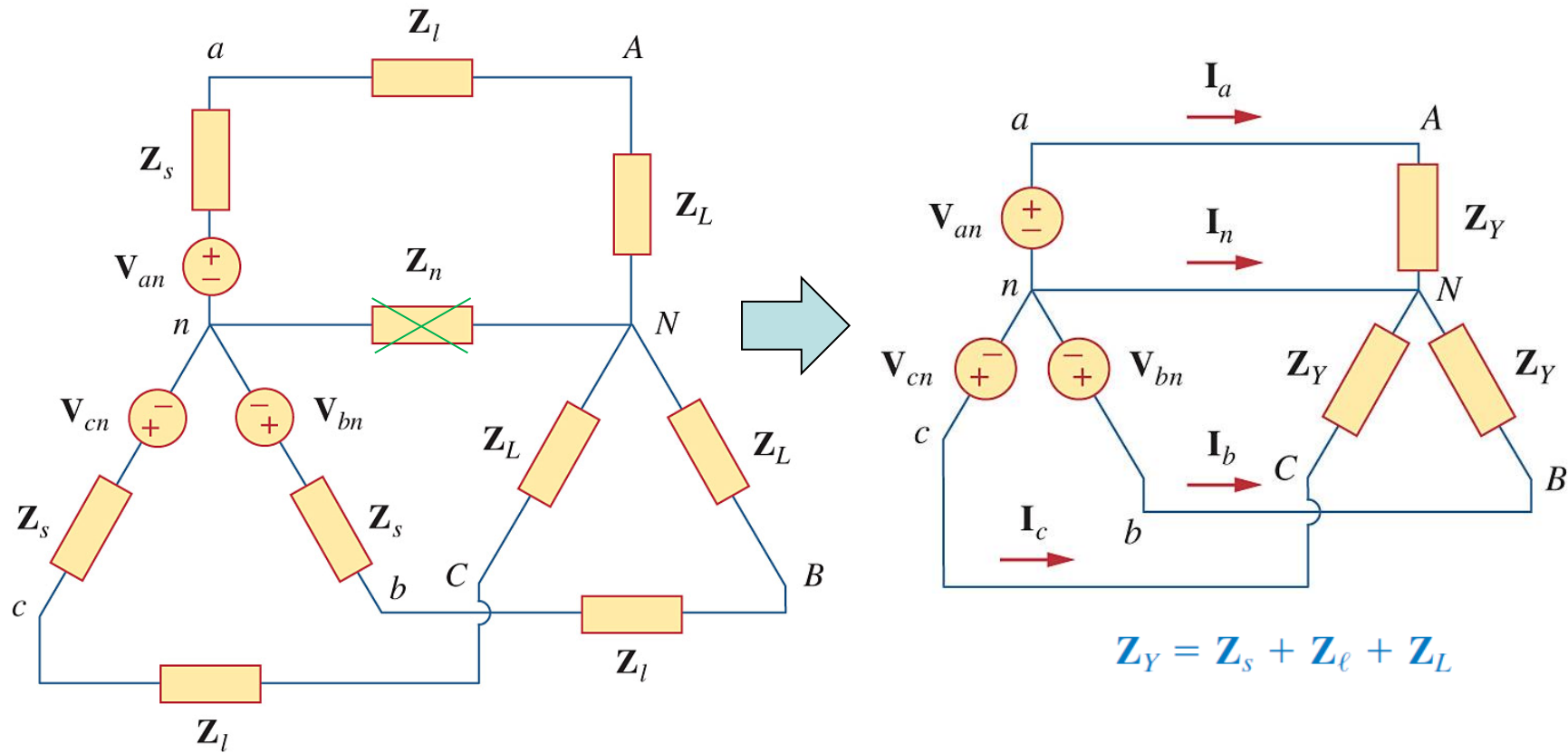
1. Y-Y connection (i.e., Y-connected source with a Y-connected load).
2. Y- Δ connection.
3. Δ - Δ connection.
4. Δ -Y connection.

12.3 Balanced Wye-Wye Connection

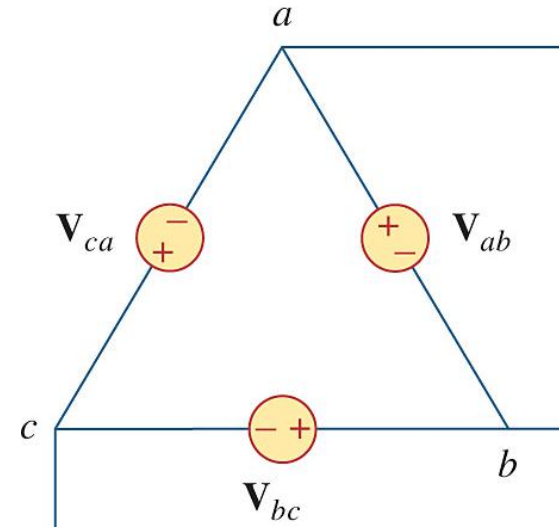
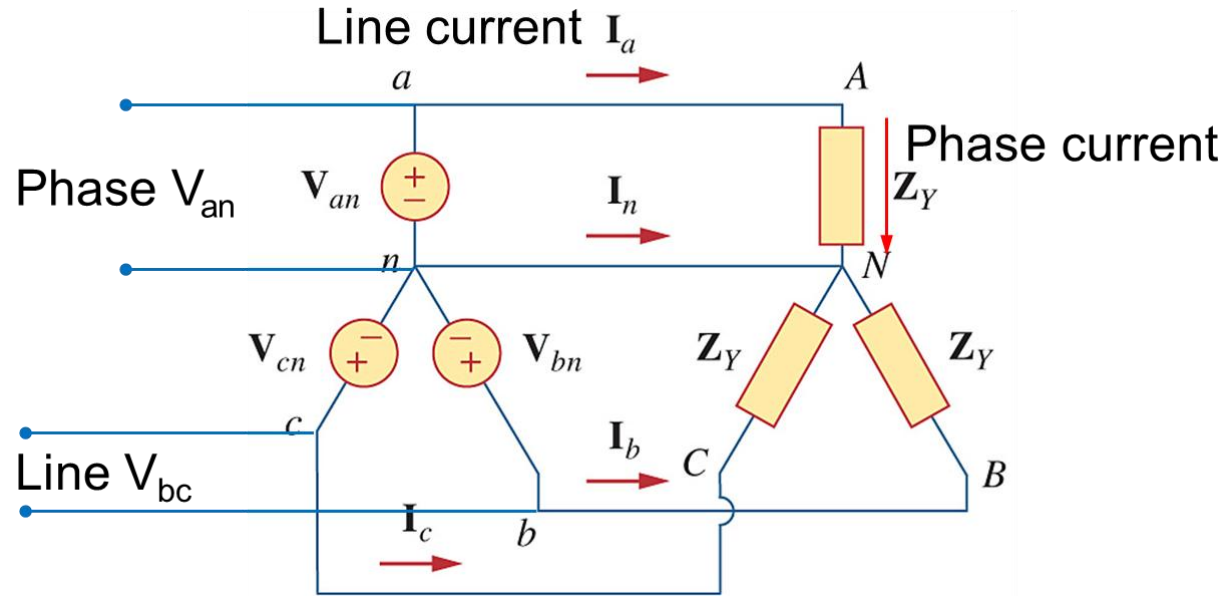
We begin with the Y-Y system, because any balanced three-phase system can be reduced to an equivalent Y-Y system. A balanced Y-Y system is a three-phase system with a balanced Y-connected source and a balanced Y-connected load.



If balanced, the system can be simplified.



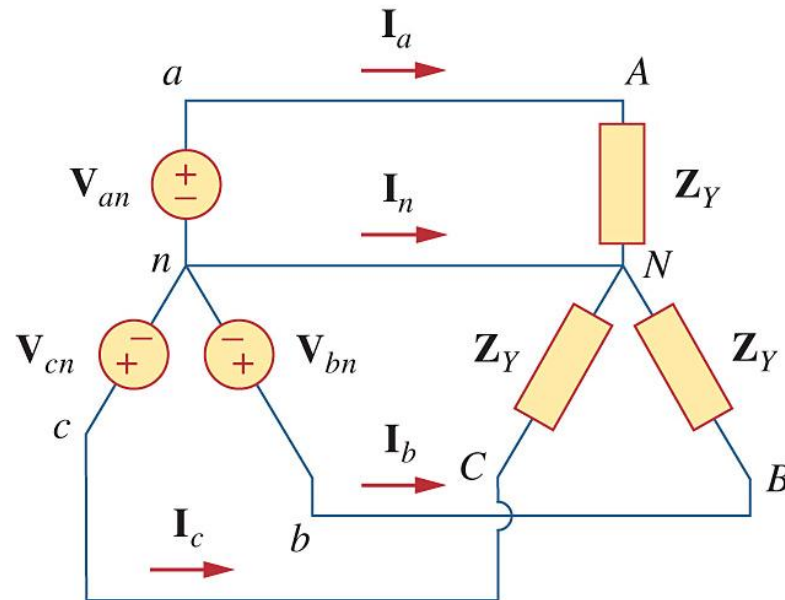
Phases and Lines?



The **voltage** between any **two** line conductors are termed **line voltages** and the **current** that passes along **each** line conductor is termed a **line current**.

Phases are connected between any pair of line terminals. The **voltage** appearing across any **phase** is termed a **phase voltage**, and the **current** passing through any **phase** is termed a **phase current**.

Assuming the positive sequence, **the phase voltages** (or line-to-neutral voltages) are



$$\tilde{V}_{an} = V_p \angle 0^\circ$$

$$\tilde{V}_{bn} = V_p \angle -120^\circ$$

$$\tilde{V}_{cn} = V_p \angle -240^\circ$$

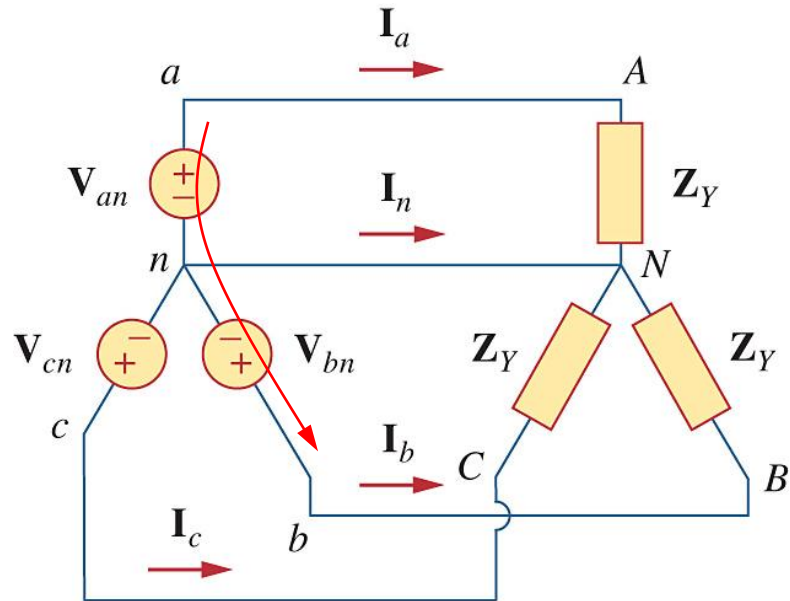
Balanced source:

Same amplitude and frequency

Out of phase with each other by 120°

Line voltages

The line voltages (or line-to-line voltages) are

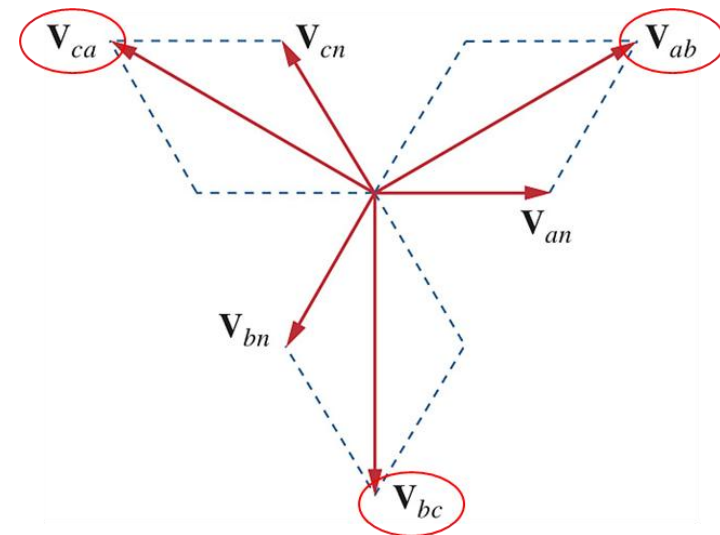


$$\tilde{V}_{ab} = \tilde{V}_{an} - \tilde{V}_{bn} = \sqrt{3}V_p \angle 30^\circ$$

$$\tilde{V}_{bc} = \tilde{V}_{bn} - \tilde{V}_{cn} = \sqrt{3}V_p \angle -90^\circ = \tilde{V}_{ab} \angle -120^\circ$$

$$\tilde{V}_{ca} = \tilde{V}_{cn} - \tilde{V}_{an} = \sqrt{3}V_p \angle -210^\circ = \tilde{V}_{ab} \angle -240^\circ$$

$$\tilde{V}_{ab} + \tilde{V}_{bc} + \tilde{V}_{ca} = 0$$

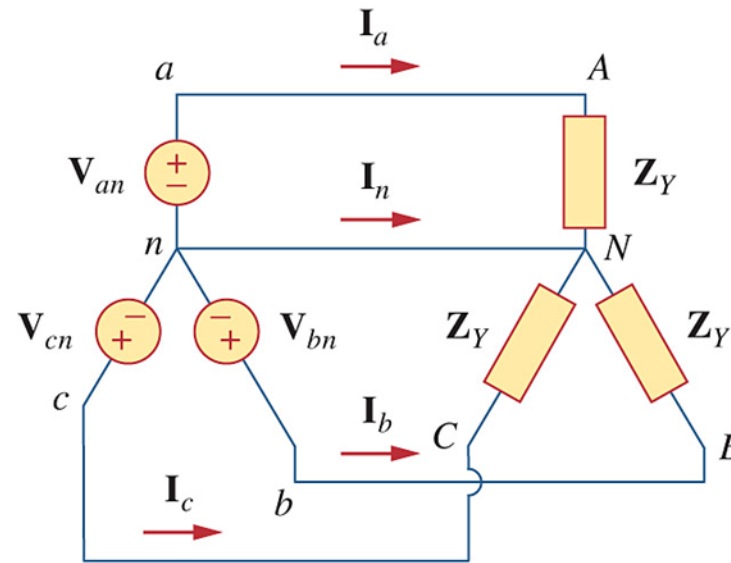


Line voltages are also balanced.

The magnitude of the line voltages is $\sqrt{3}$ times the magnitude of the phase voltages.

$$V_L = \sqrt{3}V_p$$

$$V_p = |\tilde{V}_{an}| = |\tilde{V}_{bn}| = |\tilde{V}_{cn}|$$
$$V_L = |\tilde{V}_{ab}| = |\tilde{V}_{bc}| = |\tilde{V}_{ca}|$$

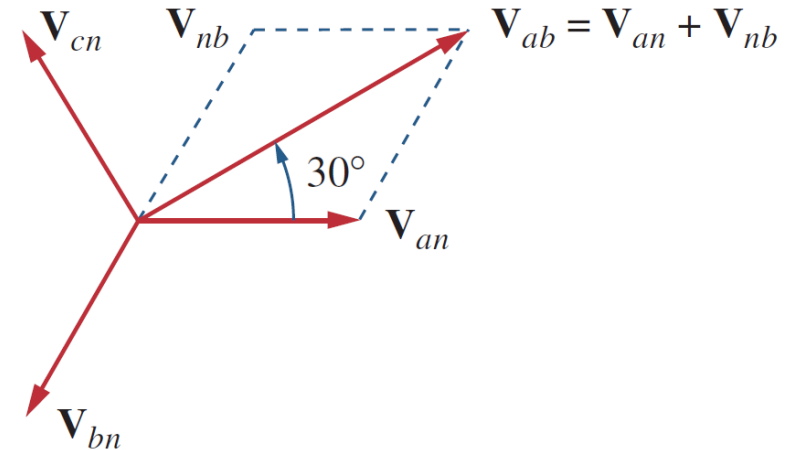


The line voltages lead their corresponding phase voltages by 30°

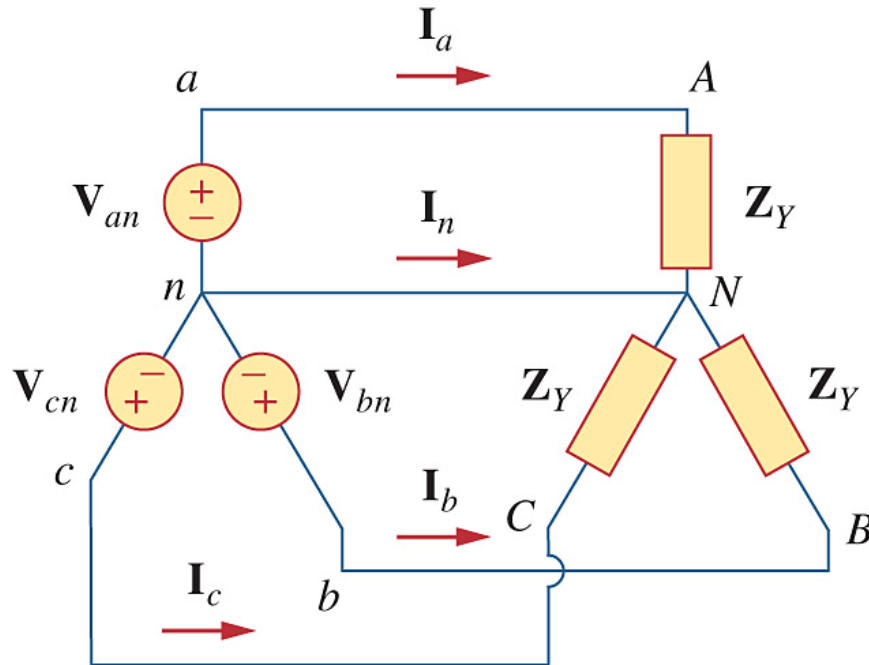
$$\begin{aligned}\tilde{V}_{an} &= V_p \angle 0^\circ & \tilde{V}_{ab} &= \tilde{V}_{an} - \tilde{V}_{bn} = \sqrt{3}V_p \angle 30^\circ \\ \tilde{V}_{bn} &= V_p \angle -120^\circ & \tilde{V}_{bc} &= \tilde{V}_{bn} - \tilde{V}_{cn} = \sqrt{3}V_p \angle -90^\circ \\ \tilde{V}_{cn} &= V_p \angle -240^\circ & \tilde{V}_{ca} &= \tilde{V}_{cn} - \tilde{V}_{an} = \sqrt{3}V_p \angle -210^\circ\end{aligned}$$

➡

$$\begin{aligned}\angle \tilde{V}_{ab} &= \angle \tilde{V}_{an} + 30^\circ \\ \angle \tilde{V}_{bc} &= \angle \tilde{V}_{bn} + 30^\circ \\ \angle \tilde{V}_{ca} &= \angle \tilde{V}_{cn} + 30^\circ\end{aligned}$$



Line currents



By KVL at each phase

$$\mathbf{I}_a = \frac{\mathbf{V}_{an}}{\mathbf{Z}_Y},$$

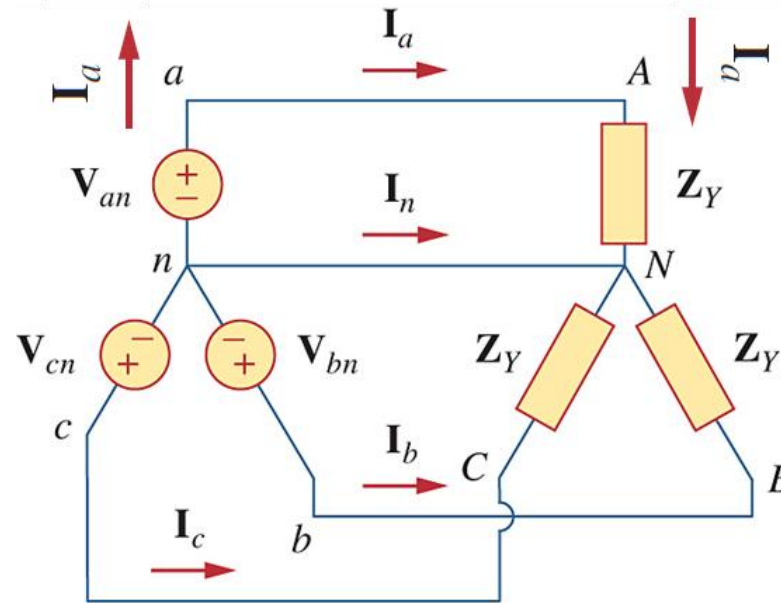
$$\mathbf{I}_b = \frac{\mathbf{V}_{bn}}{\mathbf{Z}_Y} = \frac{\mathbf{V}_{an} \angle -120^\circ}{\mathbf{Z}_Y} = \mathbf{I}_a \angle -120^\circ$$

$$\mathbf{I}_c = \frac{\mathbf{V}_{cn}}{\mathbf{Z}_Y} = \frac{\mathbf{V}_{an} \angle -240^\circ}{\mathbf{Z}_Y} = \mathbf{I}_a \angle -240^\circ$$

Line currents add up to zero: $\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c = 0$

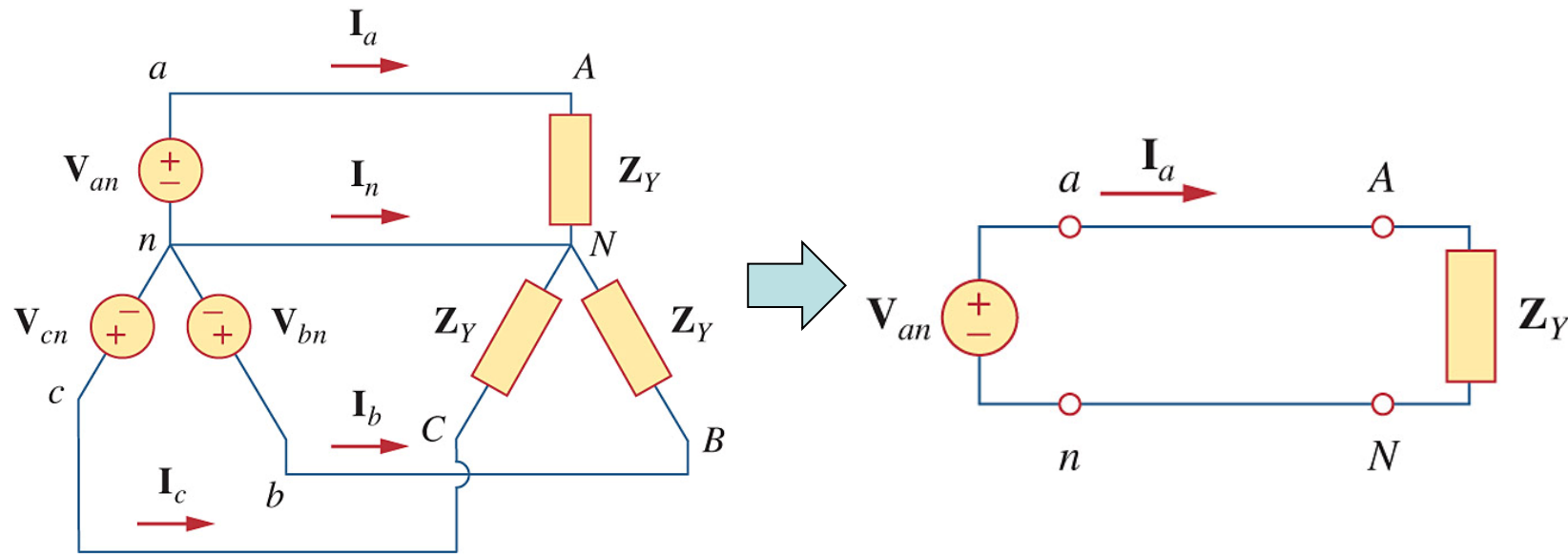
Thus, $\mathbf{I}_n = -(\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c) = 0$ and $\mathbf{V}_{nN} = \mathbf{Z}_n \mathbf{I}_n = 0$

While **the line current** is the current in each line, **the phase current** is the current in each phase of the source or load.



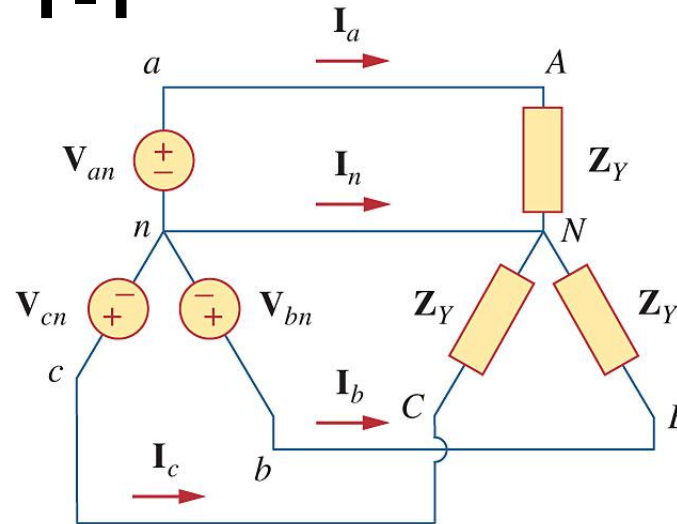
In Y-Y connected system, the line current is the same as the phase current.

An alternative way of analyzing a **balanced Y-Y system** is to do so on a **per phase basis**.



The single-phase analysis yields the line current
$$I_a = \frac{V_{an}}{Z_Y}$$

Summary of balanced Y-Y



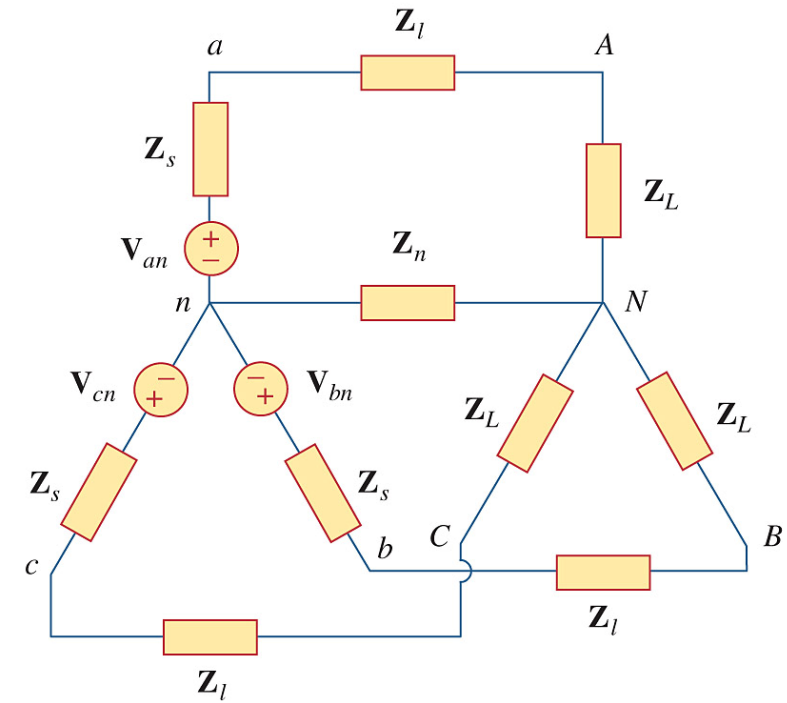
Summary of phase and line voltages/currents for balanced three-phase systems.¹

Connection	Phase voltages/currents	Line voltages/currents
Y-Y	$V_{an} = V_p \angle 0^\circ$ $V_{bn} = V_p \angle -120^\circ$ $V_{cn} = V_p \angle +120^\circ$ Same as line currents	$V_{ab} = \sqrt{3}V_p \angle 30^\circ$ $V_{bc} = V_{ab} \angle -120^\circ$ $V_{ca} = V_{ab} \angle +120^\circ$ $I_a = V_{an} / Z_Y$ $I_b = I_a \angle -120^\circ$ $I_c = I_a \angle +120^\circ$

Positive or abc sequence is assumed.

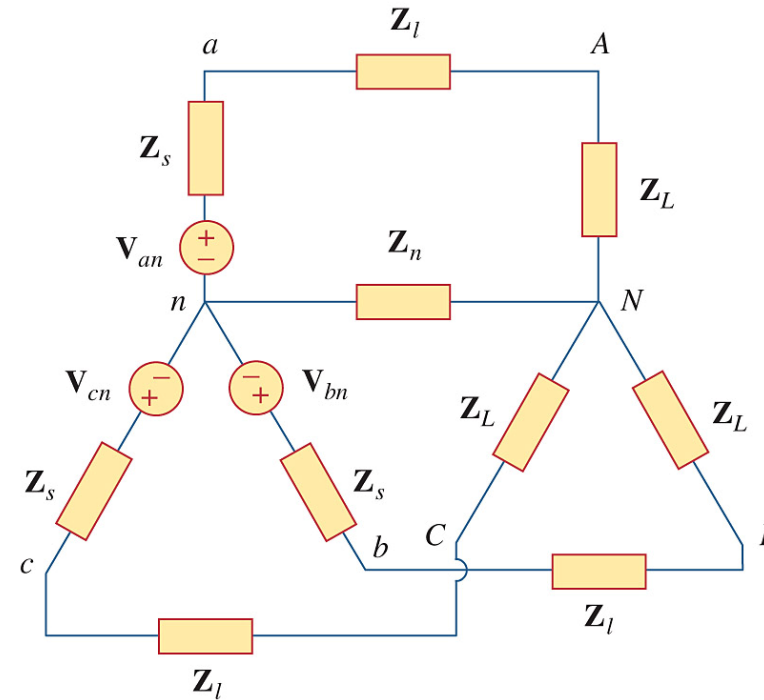
Practice Problem 12.2 A Y-connected balanced three-phase generator with an impedance of $0.4 + j0.3 \, \Omega$ per phase is connected to a Y-connected balanced load with an impedance of $24 + j19 \, \Omega$ per phase. The line joining the generator and the load has an impedance of $0.6 + j0.7 \, \Omega$ per phase. Assuming a positive sequence for the source voltages and that $\tilde{V}_{an} = 120 \angle 30^\circ \text{ V}$, find the line voltages and the line currents.

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Line voltages by follow the definition

Practice Problem 12.2 A Y-connected balanced three-phase generator with an impedance of $0.4 + j0.3 \, \Omega$ per phase is connected to a Y-connected balanced load with an impedance of $24 + j19 \, \Omega$ per phase. The line joining the generator and the load has an impedance of $0.6 + j0.7 \, \Omega$ per phase. Assuming a positive sequence for the source voltages and that $\tilde{V}_{an} = 120 \angle 30^\circ \text{ V}$, find the line voltages and the line currents.



The "ideal" line voltages are

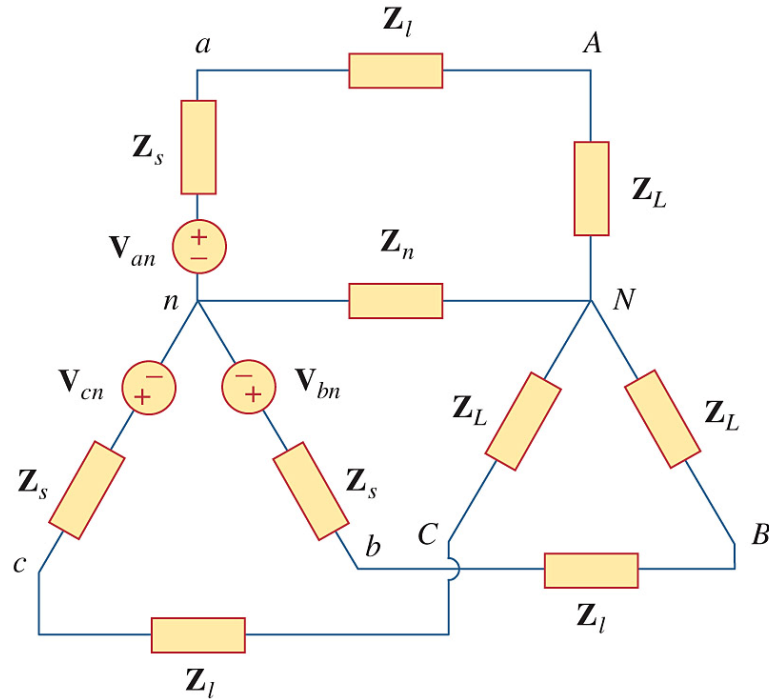
$$\tilde{V}_{ab} = \tilde{V}_{an} \sqrt{3} \angle 30^\circ = 120 \angle 30^\circ \times \sqrt{3} \angle 30^\circ$$

$$\approx 207.8461 \angle 60^\circ \text{ (V)}$$

$$\tilde{V}_{bc} = \tilde{V}_{ab} \angle -120^\circ = 207.8461 \angle -60^\circ \text{ (V)}$$

$$\tilde{V}_{ba} = \tilde{V}_{ab} \angle -240^\circ = 207.8461 \angle -180^\circ \text{ (V)}$$

Line currents



$$Z_Y = Z_s + Z_l + Z_L = (0.4 + j0.3) + (24 + j19) + (0.6 + j0.7) = 25 + j20 \text{ } (\Omega)$$

$$\tilde{I}_a = \frac{\tilde{V}_{an}}{Z_Y} = \frac{120 \angle 30^\circ}{25 + j20} \approx \frac{120 \angle 30^\circ}{32.0156 \angle 38.66^\circ}$$

$$\approx 3.7482 \angle -8.66^\circ \text{ (A)}$$

$$\tilde{I}_b = \tilde{I}_a \angle -120^\circ = 3.7482 \angle -128.66^\circ \text{ (A)}$$

$$\tilde{I}_c = \tilde{I}_a \angle -240^\circ = 3.7482 \angle -248.66^\circ \text{ (A)}$$

Alternatively, by the equations

The "ideal" line voltages are

$$\tilde{V}_{ab} = \tilde{V}_{an} \sqrt{3} \angle 30^\circ = 120 \angle 30^\circ \times \sqrt{3} \angle 30^\circ$$

$$\approx 207.8461 \angle 60^\circ \text{ (V)}$$

$$\tilde{V}_{bc} = \tilde{V}_{ab} \angle -120^\circ = 207.8461 \angle -60^\circ \text{ (V)}$$

$$\tilde{V}_{ba} = \tilde{V}_{ab} \angle -240^\circ = 207.8461 \angle -180^\circ \text{ (V)}$$

Note that for a balanced system, the other two voltages/currents can be quickly obtained by their **phase relation**.

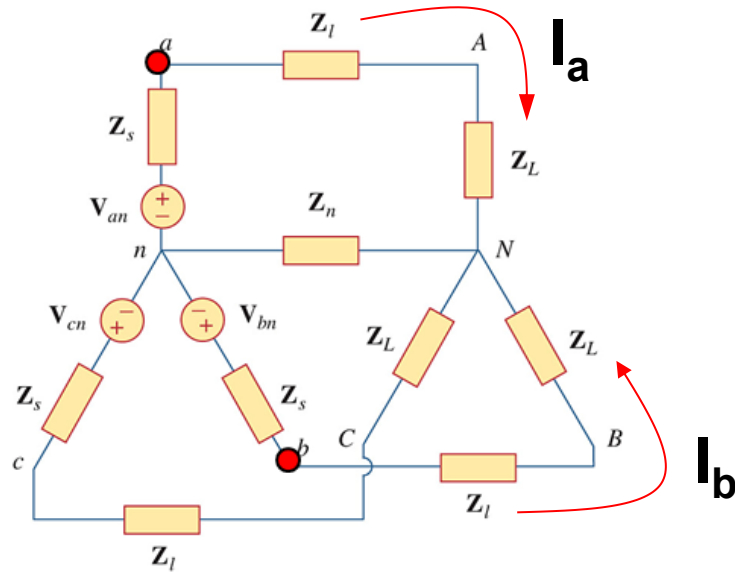
Line current

$$Z_Y = Z_s + Z_l + Z_L = (0.4 + j0.3) + (24 + j19) \\ + (0.6 + j0.7) = 25 + j20 \text{ } (\Omega)$$

$$\tilde{I}_a = \frac{\tilde{V}_{an}}{Z_Y} = \frac{120 \angle 30^\circ}{25 + j20} \approx \frac{120 \angle 30^\circ}{32.0156 \angle 38.66^\circ} \\ \approx 3.7482 \angle -8.66^\circ \text{ (A)}$$

$$\tilde{I}_b = \tilde{I}_a \angle -120^\circ = 3.7482 \angle -128.66^\circ \text{ (A)}$$

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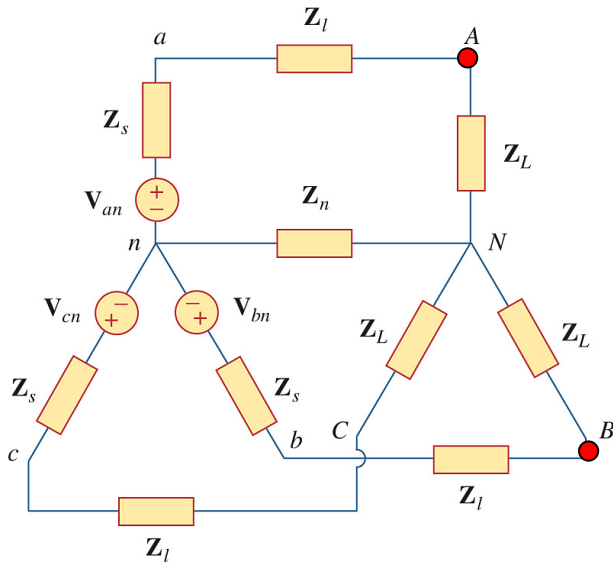


The line voltages seen from the generator terminals are

$$\begin{aligned}
 \tilde{V}_{ab} &= \tilde{I}_a(Z_l + Z_L) - \tilde{I}_b(Z_l + Z_L) \\
 &= \tilde{I}_a \sqrt{3} \angle 30^\circ (Z_l + Z_L) \\
 &= \frac{\tilde{V}_{an}}{Z_Y} \sqrt{3} \angle 30^\circ (Z_l + Z_L) = \tilde{V}_{ab} \left(\frac{Z_l + Z_L}{Z_Y} \right) \text{ Line voltage } V_{ab}
 \end{aligned}$$

$$\begin{aligned}
&= 207.8461 \angle 60^\circ \times \left(\frac{24.6 + j19.7}{25 + j20} \right) \\
&\approx 207.8461 \angle 60^\circ \times \frac{31.5159 \angle 38.69^\circ}{32.0156 \angle 38.66^\circ} \\
&\approx 204.6020 \angle 60.03^\circ \text{ (V)} \\
\tilde{V}_{bc} &= 204.6020 \angle -59.97^\circ \text{ (V)} \\
\tilde{V}_{ca} &= 204.6020 \angle -179.97^\circ \text{ (V)}
\end{aligned}$$

Slightly lower magnitude and angles compared to the ideal line voltages.



The line voltages seen from the load terminals are

$$\tilde{V}_{AB} = \tilde{I}_a Z_L - \tilde{I}_b Z_L$$

$$I_a = 3.7482 \angle -8.66$$

$$I_b = 3.7482 \angle -8.66$$

$$Z_L = 14 + j19 = 30.61 \angle 38.37$$

$$V_{AB} = 198.72 \angle 59.71$$

$$V_{BC} = 198.72 \angle -60.29$$

$$V_{CA} = 198.72 \angle -180.29$$

The line voltages seen from the load terminals are

$$\tilde{V}_{AB} = \tilde{I}_a Z_L - \tilde{I}_b Z_L = \tilde{V}_{ab} \frac{Z_L}{Z_Y}$$

$\tilde{V}_{AN} - \tilde{V}_{BN}$ $I_a - I_b = I_a \sqrt{3} \angle 30^\circ = V_{an} / Z_Y \sqrt{3} \angle 30^\circ = V_{ab} / Z_Y$

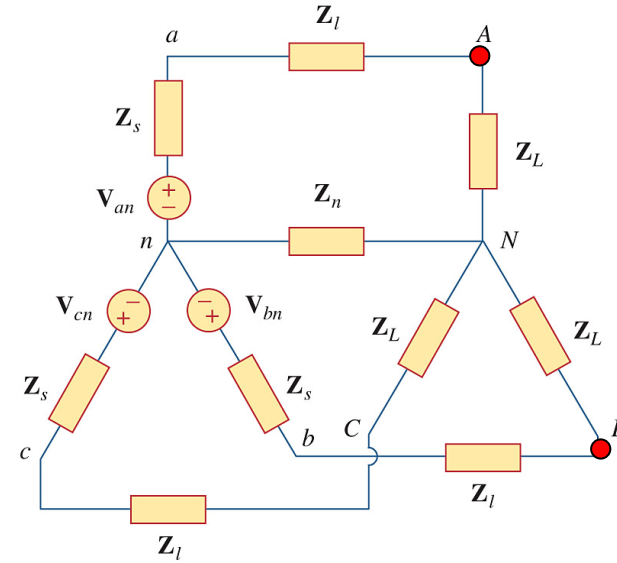
$$= 207.8461 \angle 60^\circ \times \left(\frac{24 + 19}{25 + j20} \right)$$

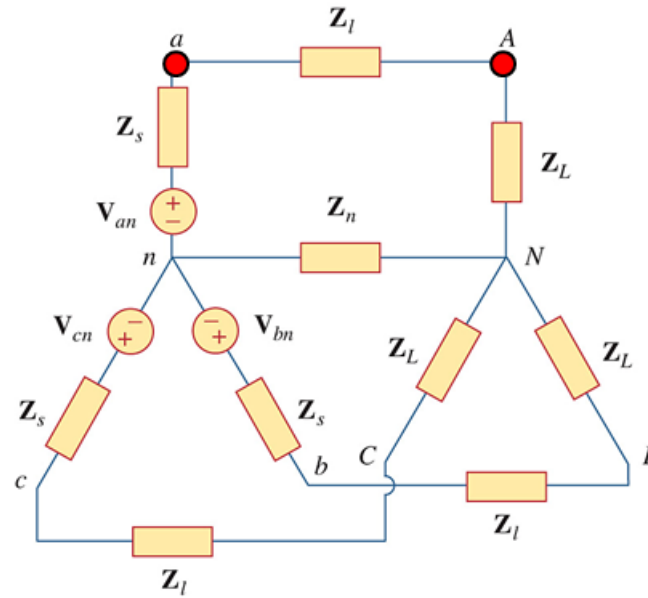
$$\approx 207.8461 \angle 60^\circ \times \frac{30.6105 \angle 38.37^\circ}{32.0156 \angle 38.66^\circ}$$

$$\approx 198.7242 \angle 59.71^\circ \text{ (V)}$$

$$\tilde{V}_{BC} \approx 198.7242 \angle -60.29^\circ \text{ (V)}$$

$$\tilde{V}_{CA} \approx 198.7242 \angle -180.29^\circ \text{ (V)}$$





The line voltage drops are

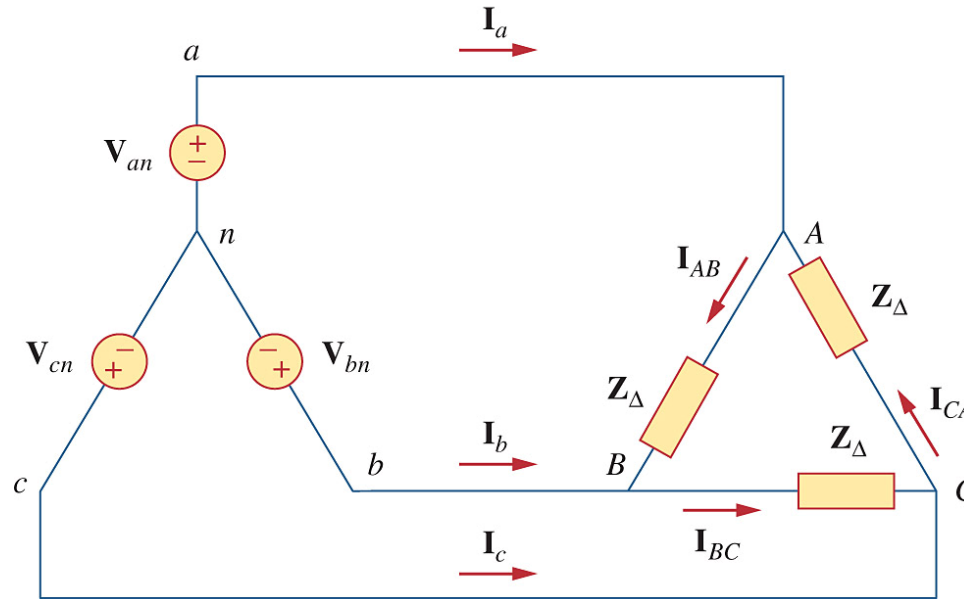
$$\tilde{I}_a Z_l = 3.7482 \angle -8.66^\circ \times (0.6 + j0.7)$$

$$\approx 3.4558 \angle 40.74^\circ \text{ (V)}$$

$$\tilde{I}_b Z_l = 3.4558 \angle -79.26^\circ \text{ (V)}$$

$$\tilde{I}_c Z_l = 3.4558 \angle -199.26^\circ \text{ (V)}$$

12.4 Balanced Wye-Delta Connection



A balanced Y-Δ system consists of a balanced Y-connected source feeding a balanced Δ-connected load.

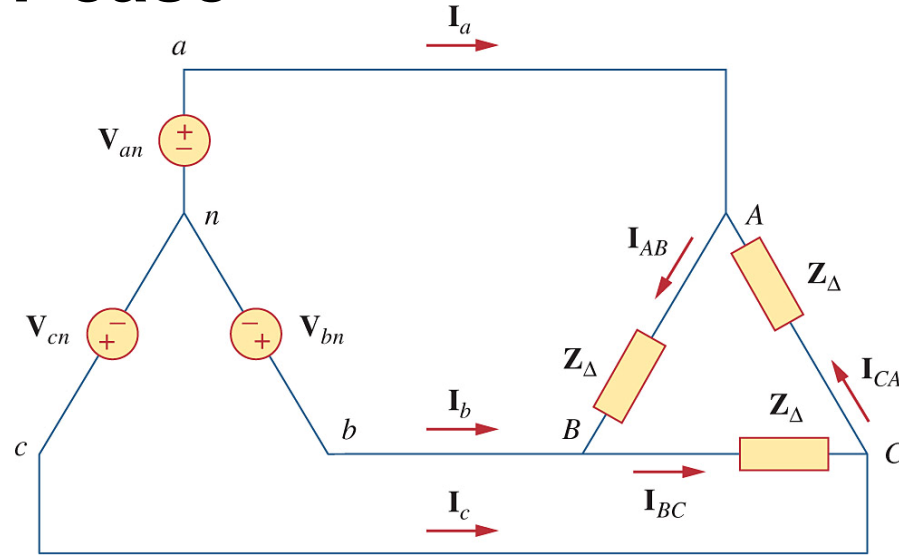
Phase voltages: same as Y-Y case

Assume that

$$\tilde{V}_{an} = V_p \angle 0^\circ$$

$$\tilde{V}_{bn} = V_p \angle -120^\circ$$

$$\tilde{V}_{cn} = V_p \angle -240^\circ$$



The line voltages are

$$\tilde{V}_{ab} = \sqrt{3}V_p \angle 30^\circ = \tilde{V}_{an}\sqrt{3} \angle 30^\circ = \tilde{V}_{AB}$$

$$\tilde{V}_{bc} = \tilde{V}_{ab} \angle -120^\circ = \tilde{V}_{BC}$$

$$\tilde{V}_{ca} = \tilde{V}_{ab} \angle -240^\circ = \tilde{V}_{CA}$$

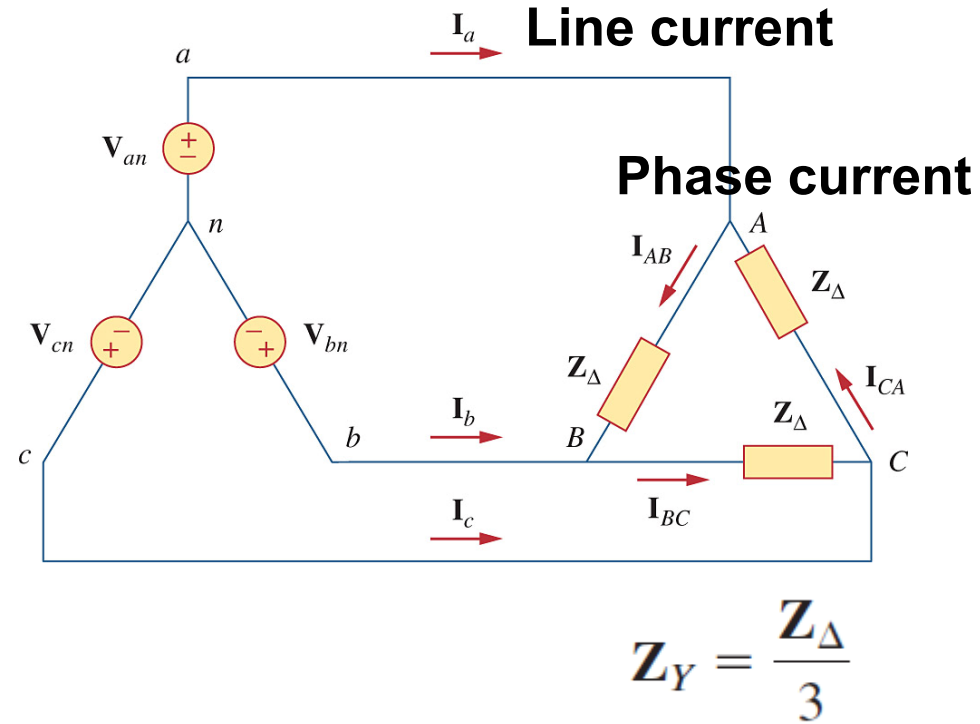
Phase currents

The phase currents are

$$\tilde{I}_{AB} = \frac{\tilde{V}_{AB}}{Z_{\Delta}}$$

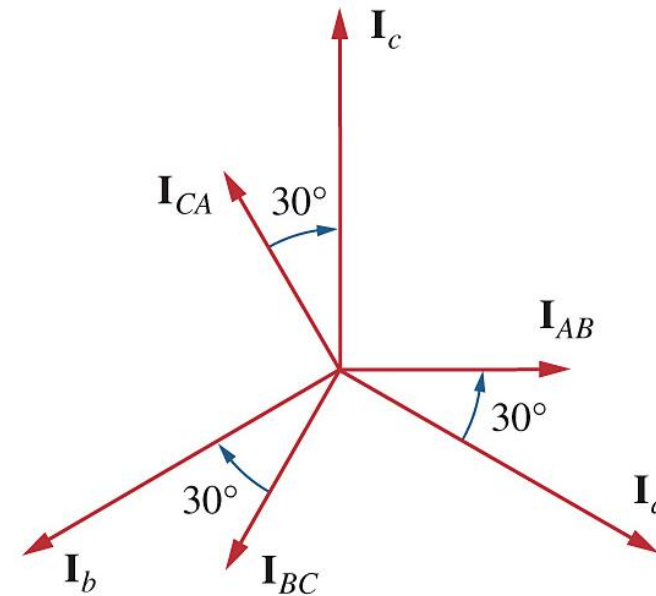
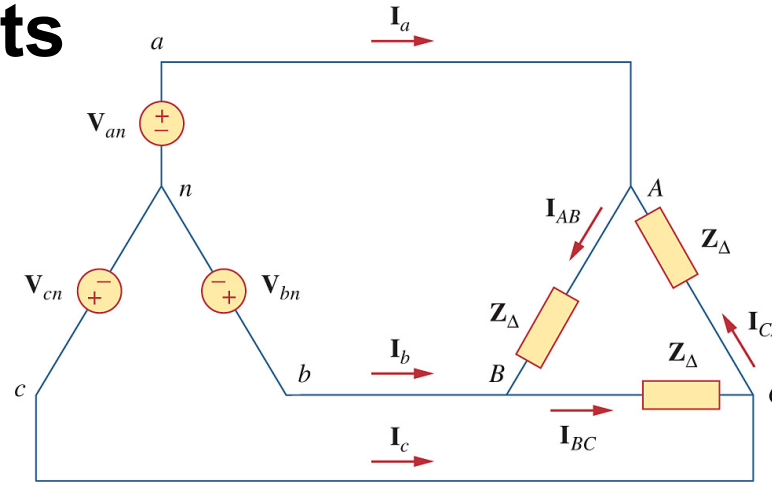
$$\tilde{I}_{BC} = \frac{\tilde{V}_{BC}}{Z_{\Delta}} = \tilde{I}_{AB} \angle -120^{\circ}$$

$$\tilde{I}_{CA} = \frac{\tilde{V}_{CA}}{Z_{\Delta}} = \tilde{I}_{AB} \angle -240^{\circ}$$



While the **line current** is the current in each line, the **phase current** is the current in each phase of the source or load.

Line currents



The line currents are

$$\tilde{I}_a = \tilde{I}_{AB} - \tilde{I}_{CA} = \tilde{I}_{AB} \sqrt{3} \angle -30^\circ$$

$$\tilde{I}_b = \tilde{I}_{BC} - \tilde{I}_{AB} = \tilde{I}_{BC} \sqrt{3} \angle -30^\circ = \tilde{I}_a \angle -120^\circ$$

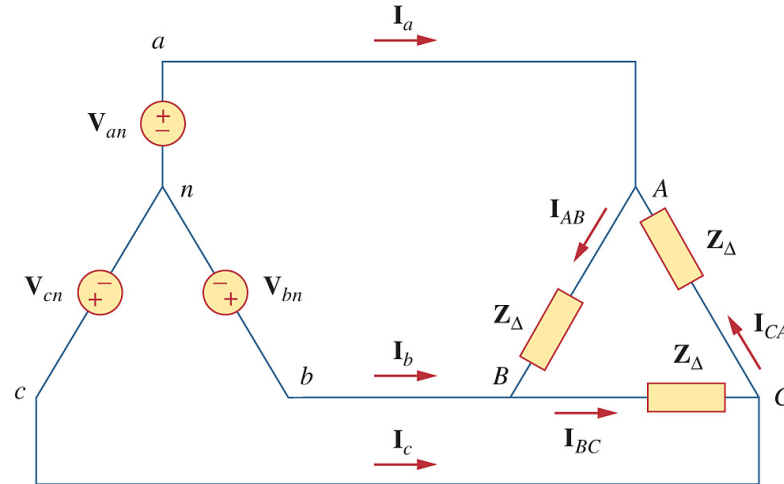
$$\tilde{I}_c = \tilde{I}_{CA} - \tilde{I}_{BC} = \tilde{I}_{CA} \sqrt{3} \angle -30^\circ = \tilde{I}_a \angle -240^\circ$$

showing that

$$I_L = \sqrt{3} I_p \quad I_L = |\tilde{I}_a| = |\tilde{I}_b| = |\tilde{I}_c|, \quad I_p = |\tilde{I}_{AB}| = |\tilde{I}_{BC}| = |\tilde{I}_{CA}|$$

$$\begin{aligned} \tilde{I}_{AB} &= \frac{\tilde{V}_{AB}}{Z_\Delta} \\ \tilde{I}_{BC} &= \frac{\tilde{V}_{BC}}{Z_\Delta} = \tilde{I}_{AB} \angle -120^\circ \\ \tilde{I}_{CA} &= \frac{\tilde{V}_{CA}}{Z_\Delta} = \tilde{I}_{AB} \angle -240^\circ \end{aligned}$$

Summary of balanced Y-Δ

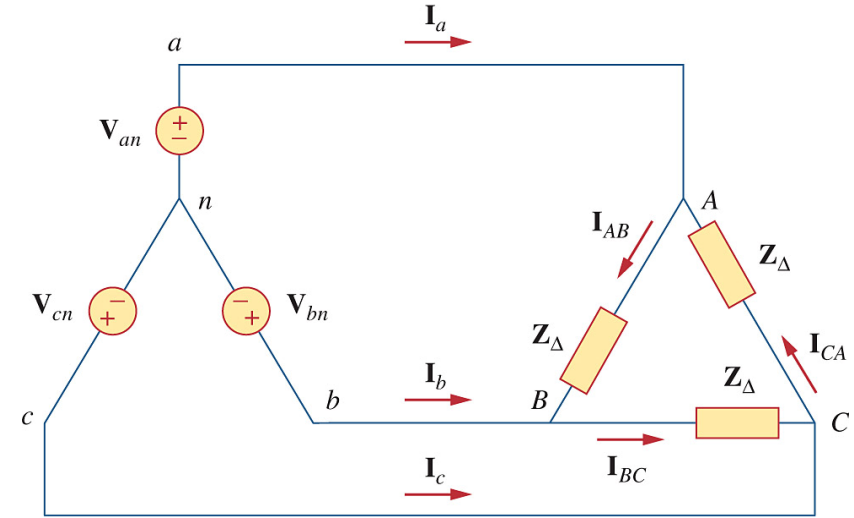


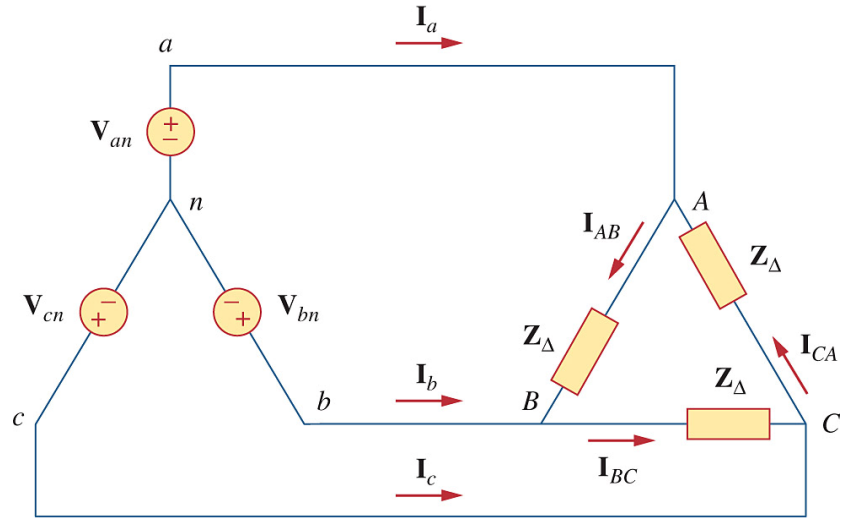
Summary of phase and line voltages/currents for balanced three-phase systems.¹

Connection	Phase voltages/currents	Line voltages/currents
Y-Δ	$\begin{aligned} \mathbf{V}_{an} &= V_p \angle 0^\circ \\ \mathbf{V}_{bn} &= V_p \angle -120^\circ \\ \mathbf{V}_{cn} &= V_p \angle +120^\circ \\ \mathbf{I}_{AB} &= \mathbf{V}_{AB} / \mathbf{Z}_\Delta \\ \mathbf{I}_{BC} &= \mathbf{V}_{BC} / \mathbf{Z}_\Delta \\ \mathbf{I}_{CA} &= \mathbf{V}_{CA} / \mathbf{Z}_\Delta \end{aligned}$	$\begin{aligned} \mathbf{V}_{ab} &= \mathbf{V}_{AB} = \sqrt{3} V_p \angle 30^\circ \\ \mathbf{V}_{bc} &= \mathbf{V}_{BC} = \mathbf{V}_{ab} \angle -120^\circ \\ \mathbf{V}_{ca} &= \mathbf{V}_{CA} = \mathbf{V}_{ab} \angle +120^\circ \\ \mathbf{I}_a &= \mathbf{I}_{AB} \sqrt{3} \angle -30^\circ \\ \mathbf{I}_b &= \mathbf{I}_a \angle -120^\circ \\ \mathbf{I}_c &= \mathbf{I}_a \angle +120^\circ \end{aligned}$

Positive or abc sequence is assumed.

Practice Problem 12.3 One line voltage of a balanced Y-connected source is $\tilde{V}_{AB} = 240\angle -20^\circ$ V. If the source is connected to a Δ -connected load of $20\angle 40^\circ \Omega$, find the phase and line currents. Assume the abc sequence.





$$\tilde{I}_{AB} = \frac{\tilde{V}_{AB}}{Z_{\Delta}} = \frac{240 \angle -20^{\circ}}{20 \angle 40^{\circ}} = 12 \angle -60^{\circ} \text{ (A)}$$

$$\tilde{I}_{BC} = \tilde{I}_{AB} \angle -120^{\circ} = 12 \angle -180^{\circ} \text{ (A)}$$

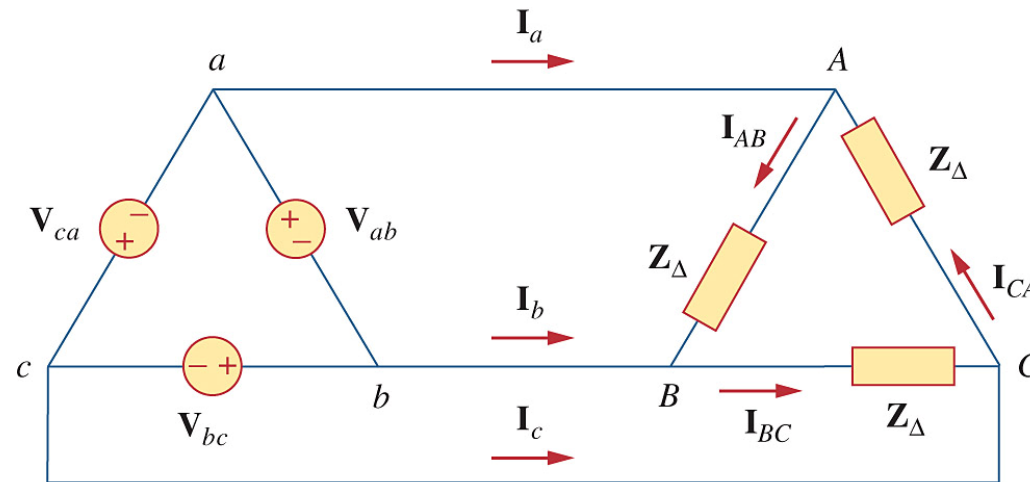
$$\tilde{I}_{CA} = \tilde{I}_{AB} \angle +120^{\circ} = 12 \angle 60^{\circ} \text{ (A)}$$

$$\tilde{I}_a = \tilde{I}_{AB} \sqrt{3} \angle -30^{\circ} \approx 20.7846 \angle -90^{\circ} \text{ (A)}$$

$$\tilde{I}_b = \tilde{I}_a \angle +240^{\circ} = 20.7846 \angle 150^{\circ} \text{ (A)}$$

$$\tilde{I}_c = \tilde{I}_a \angle +120^{\circ} = 20.7846 \angle 30^{\circ} \text{ (A)}$$

12.5 Balanced Delta-Delta Connection



A balanced Δ - Δ system is one in which both the balanced source and balanced load are Δ -connected.

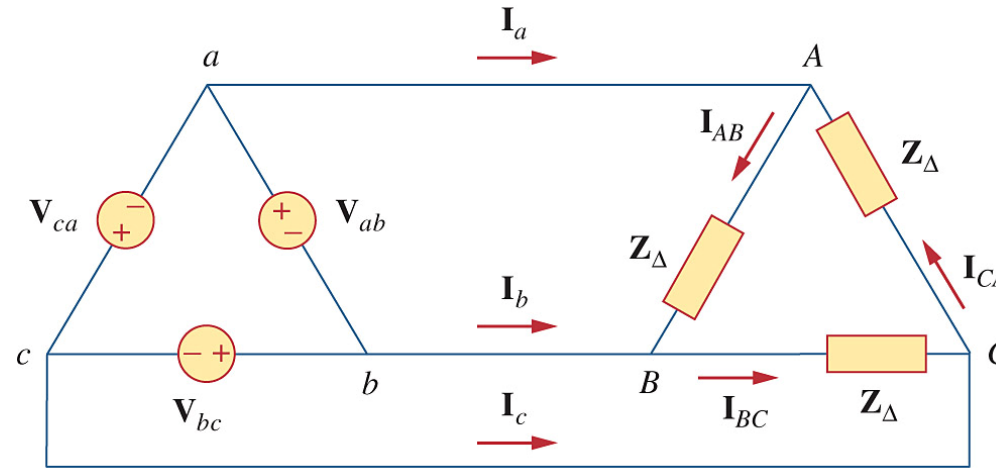
Phase and line voltages

Assume that

$$\tilde{V}_{ab} = V_p \angle 0^\circ$$

$$\tilde{V}_{bc} = V_p \angle -120^\circ$$

$$\tilde{V}_{ca} = V_p \angle +120^\circ$$

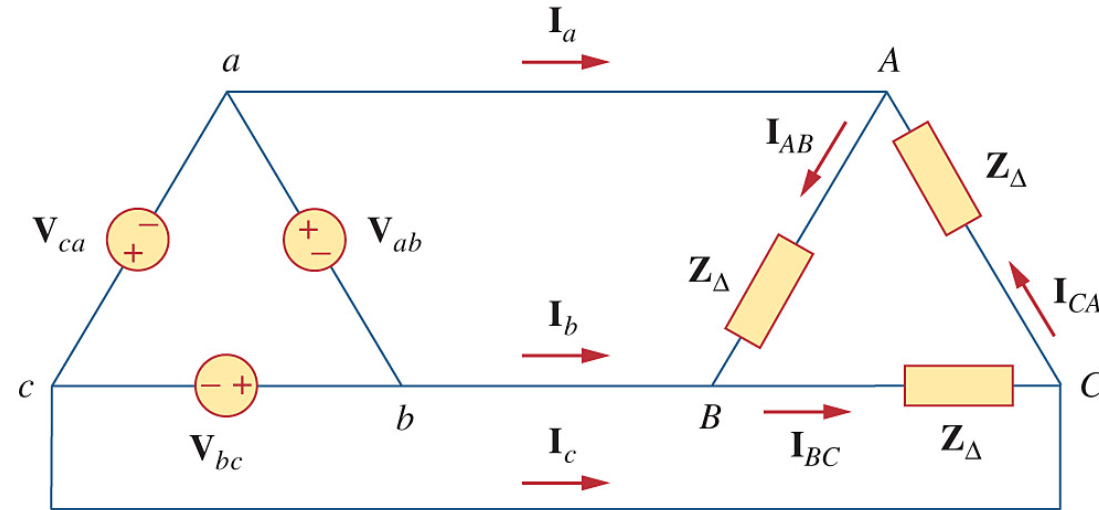


Phase voltage = Line voltage

Phase voltage = Voltage across the impedances

$$\tilde{V}_{ab} = \tilde{V}_{AB}, \tilde{V}_{bc} = \tilde{V}_{BC}, \tilde{V}_{ca} = \tilde{V}_{CA}$$

Phase current



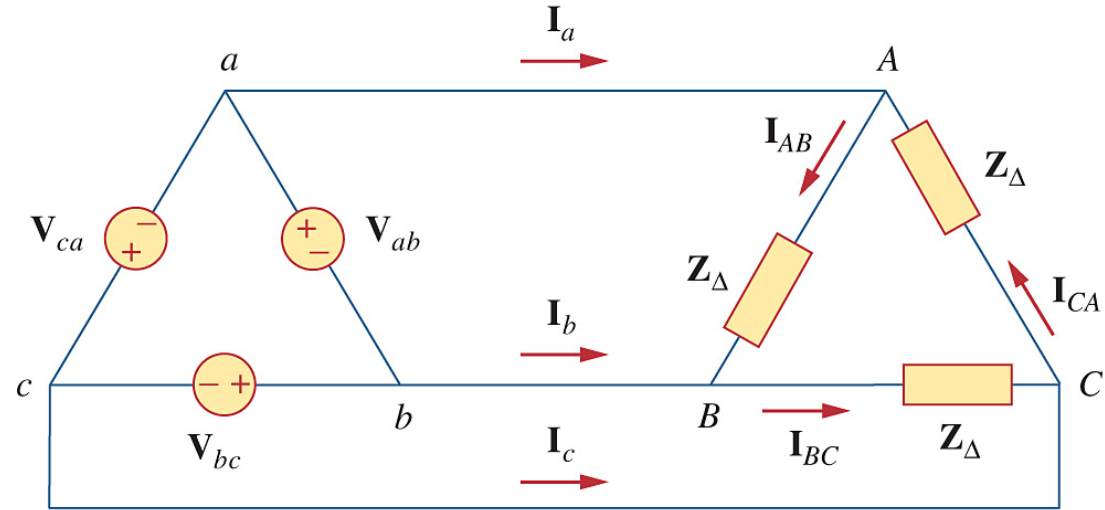
The phase currents are

$$\tilde{I}_{AB} = \frac{\tilde{V}_{AB}}{Z_{\Delta}}$$

$$\tilde{I}_{BC} = \frac{\tilde{V}_{BC}}{Z_{\Delta}} = \tilde{I}_{AB} \angle -120^{\circ}$$

$$\tilde{I}_{CA} = \frac{\tilde{V}_{CA}}{Z_{\Delta}} = \tilde{I}_{AB} \angle +120^{\circ}$$

Line current



The line currents are

$$\tilde{I}_a = \tilde{I}_{AB} - \tilde{I}_{CA} = \tilde{I}_{AB}\sqrt{3}\angle -30^\circ$$

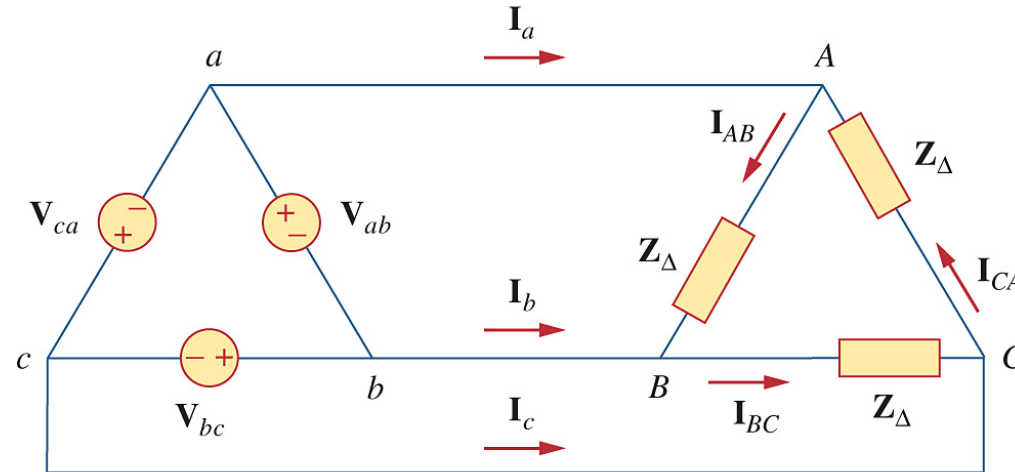
$$\tilde{I}_b = \tilde{I}_{BC} - \tilde{I}_{AB} = \tilde{I}_{BC}\sqrt{3}\angle -30^\circ = \tilde{I}_a\angle -120^\circ$$

$$\tilde{I}_c = \tilde{I}_{CA} - \tilde{I}_{BC} = \tilde{I}_{CA}\sqrt{3}\angle -30^\circ = \tilde{I}_a\angle -240^\circ$$

showing that

$$I_L = \sqrt{3}I_p \quad I_L = |\tilde{I}_a| = |\tilde{I}_b| = |\tilde{I}_c|, I_p = |\tilde{I}_{AB}| = |\tilde{I}_{BC}| = |\tilde{I}_{CA}|$$

Summary of balanced Δ - Δ

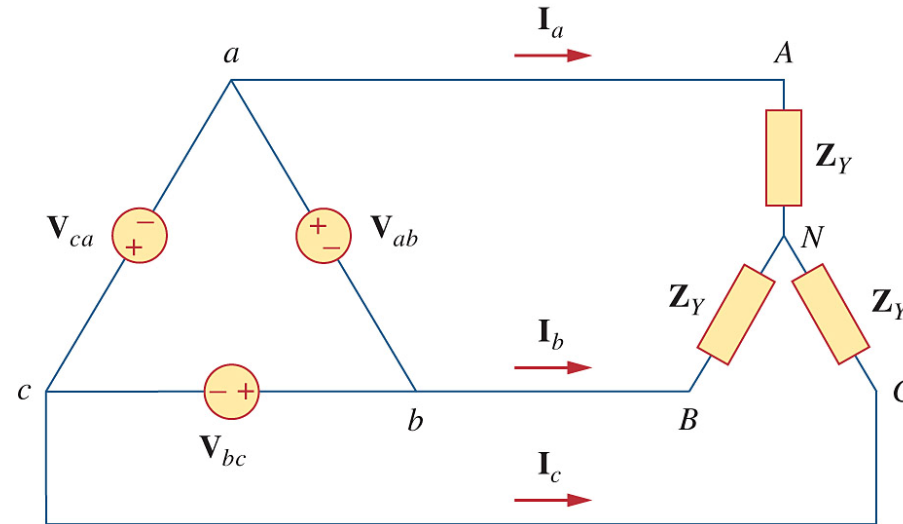


Summary of phase and line voltages/currents for balanced three-phase systems.¹

Connection	Phase voltages/currents	Line voltages/currents
Δ - Δ	$V_{ab} = V_p \angle 0^\circ$ $V_{bc} = V_p \angle -120^\circ$ $V_{ca} = V_p \angle +120^\circ$ $I_{AB} = V_{ab} / Z_{\Delta}$ $I_{BC} = V_{bc} / Z_{\Delta}$ $I_{CA} = V_{ca} / Z_{\Delta}$	Same as phase voltages $I_a = I_{AB} \sqrt{3} \angle -30^\circ$ $I_b = I_a \angle -120^\circ$ $I_c = I_a \angle +120^\circ$

Positive or abc sequence is assumed.

12.6 Balanced Delta-Wye Connection



A balanced Δ -Y system consists of a balanced Δ -connected source feeding a balanced Y-connected load.

***In Δ -connected source:**

Phase voltages = Line voltages

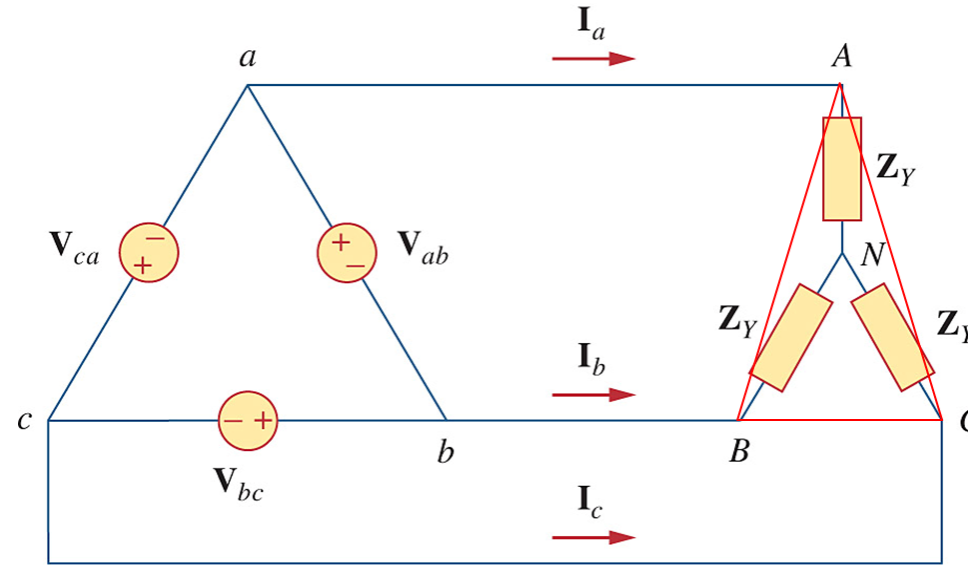
Line current

Assume that

$$\tilde{V}_{ab} = V_p \angle 0^\circ$$

$$\tilde{V}_{bc} = V_p \angle -120^\circ$$

$$\tilde{V}_{ca} = V_p \angle +120^\circ$$

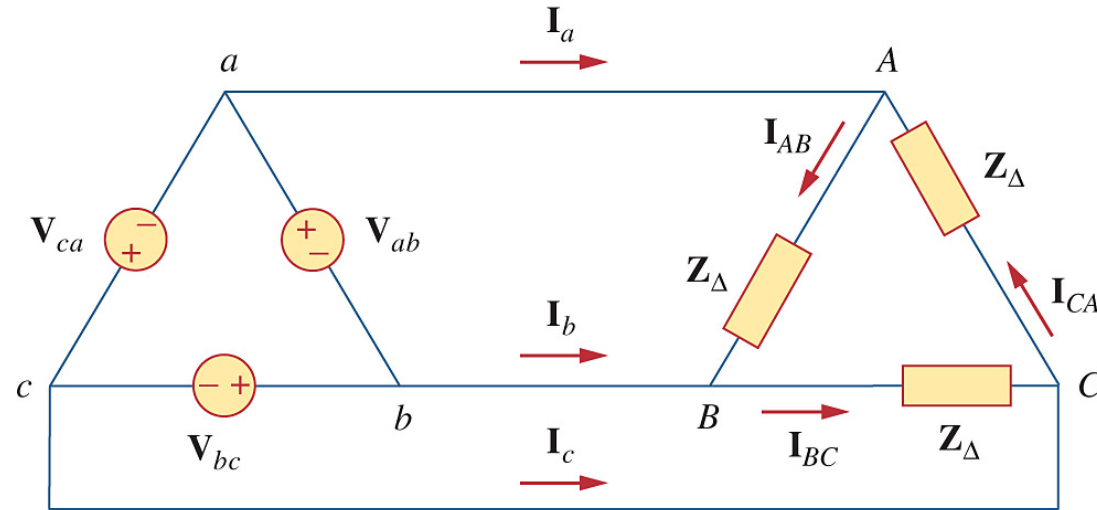


Replace the Y-connected load with its equivalent Δ -connected load. **Line current (e.g. I_a)** in a Δ -connected load is:

$$\begin{aligned} \tilde{I}_a &= \tilde{I}_{AB} \sqrt{3} \angle -30^\circ = \frac{\tilde{V}_{AB}}{Z_\Delta} \sqrt{3} \angle -30^\circ \\ &= \frac{\tilde{V}_{ab}}{\cancel{3Z_Y}} \sqrt{3} \angle -30^\circ = \frac{\tilde{V}_{ab}}{\sqrt{3}Z_Y} \angle -30^\circ \end{aligned}$$

$$\begin{aligned} Z_\Delta &= 3Z_Y \\ Z_Y &= Z_\Delta / 3 \end{aligned}$$

Recall: Δ - Δ Phase current



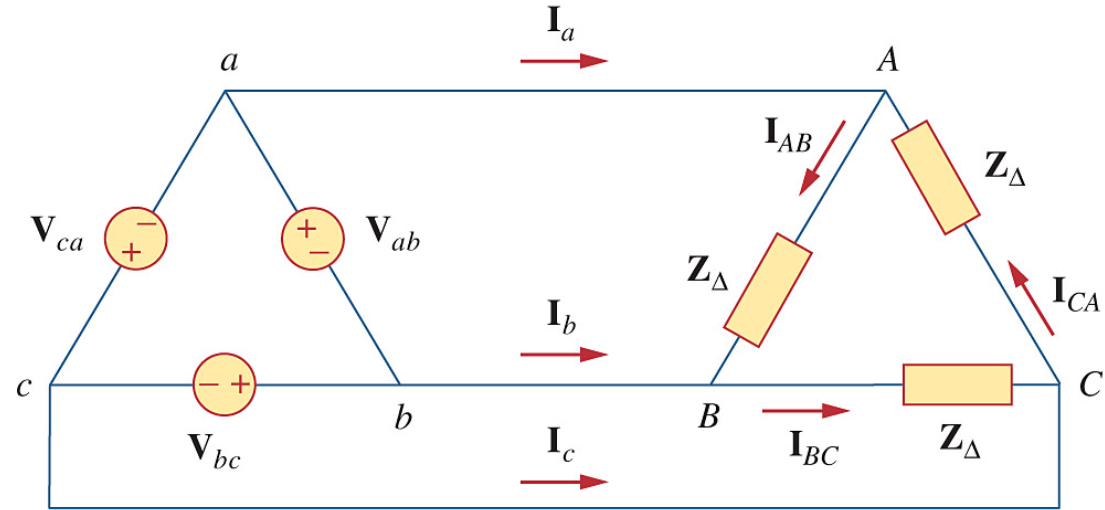
The phase currents are

$$\tilde{I}_{AB} = \frac{\tilde{V}_{AB}}{Z_{\Delta}}$$

$$\tilde{I}_{BC} = \frac{\tilde{V}_{BC}}{Z_{\Delta}} = \tilde{I}_{AB} \angle -120^{\circ}$$

$$\tilde{I}_{CA} = \frac{\tilde{V}_{CA}}{Z_{\Delta}} = \tilde{I}_{AB} \angle +120^{\circ}$$

Recall: Δ - Δ Line current



The line currents are

$$\tilde{I}_a = \tilde{I}_{AB} - \tilde{I}_{CA} = \tilde{I}_{AB} \sqrt{3} \angle -30^\circ$$

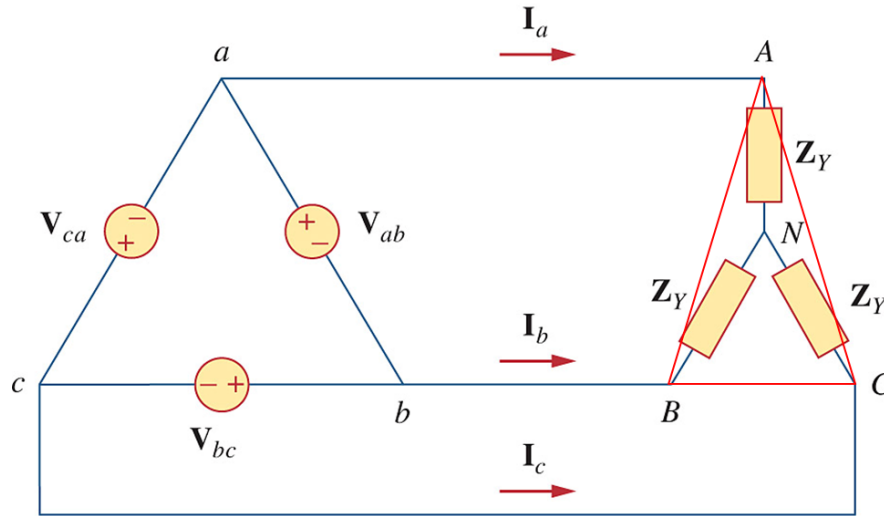
$$\tilde{I}_b = \tilde{I}_{BC} - \tilde{I}_{AB} = \tilde{I}_{BC} \sqrt{3} \angle -30^\circ = \tilde{I}_a \angle -120^\circ$$

$$\tilde{I}_c = \tilde{I}_{CA} - \tilde{I}_{BC} = \tilde{I}_{CA} \sqrt{3} \angle -30^\circ = \tilde{I}_a \angle -240^\circ$$

showing that

$$I_L = \sqrt{3} I_p \quad I_L = |\tilde{I}_a| = |\tilde{I}_b| = |\tilde{I}_c|, I_p = |\tilde{I}_{AB}| = |\tilde{I}_{BC}| = |\tilde{I}_{CA}|$$

Line current



$$\tilde{I}_a = \frac{\tilde{V}_{ab}}{\sqrt{3}Z_Y} \angle -30^\circ$$

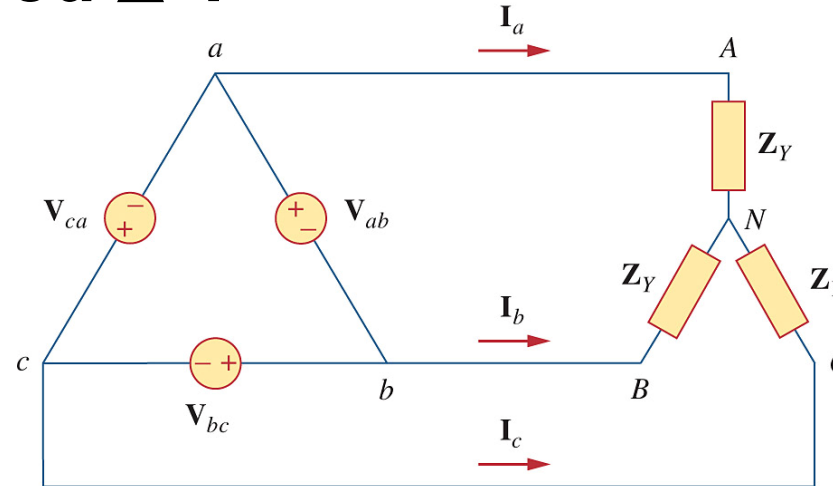
$$\tilde{I}_b = \frac{\tilde{V}_{bc}}{\sqrt{3}Z_Y} \angle -30^\circ = \tilde{I}_a \angle -120^\circ$$

$$\tilde{I}_c = \frac{\tilde{V}_{ca}}{\sqrt{3}Z_Y} \angle -30^\circ = \tilde{I}_a \angle +120^\circ$$

$$\begin{aligned}\tilde{V}_{ab} &= V_p \angle 0^\circ \\ \tilde{V}_{bc} &= V_p \angle -120^\circ \\ \tilde{V}_{ca} &= V_p \angle +120^\circ\end{aligned}$$

In **Y**-connected load,
Phase currents = Line currents

Summary of balanced Δ -Y



Summary of phase and line voltages/currents for balanced three-phase systems.¹

Connection	Phase voltages/currents	Line voltages/currents
Δ -Y	$V_{ab} = V_p \angle 0^\circ$	Same as phase voltages
	$V_{bc} = V_p \angle -120^\circ$	
	$V_{ca} = V_p \angle +120^\circ$	
	Same as line currents	$I_a = \frac{V_p \angle -30^\circ}{\sqrt{3}Z_Y}$
		$I_b = I_a \angle -120^\circ$
		$I_c = I_a \angle +120^\circ$

Positive or abc sequence is assumed.

In a balanced Δ -Y circuit, $\mathbf{V}_{ab} = 240\angle 15^\circ$ and $\mathbf{Z}_Y = (12 + j15) \Omega$.
Calculate the line currents.

Practice Problem 12.5

Answer: $7.21\angle -66.34^\circ$ A, $7.21\angle +173.66^\circ$ A, $7.21\angle 53.66^\circ$ A.

$$\begin{aligned} Z_{\Delta} &= 3Z_Y \\ Z_Y &= Z_{\Delta}/3 \end{aligned}$$

12.7 Power in a Balanced System

Three phase system - Advantage 1:

We begin by examining the **instantaneous power** absorbed by the load (time domain). For a **Y-connected load**, the phase voltages in the abc sequence are

$$v_{AN} = \sqrt{2}V_p \cos \omega t$$

$$v_{BN} = \sqrt{2}V_p \cos(\omega t - 120^\circ)$$

$$v_{CN} = \sqrt{2}V_p \cos(\omega t + 120^\circ)$$

Phase voltage V_p is the rms value

→ Peak value: $\sqrt{2}V_p$

If $Z_Y = Z \angle \theta$, the phase currents are

$$i_a = \sqrt{2} I_p \cos(\omega t - \theta)$$

$$i_b = \sqrt{2} I_p \cos(\omega t - \theta - 120^\circ)$$

$$i_c = \sqrt{2} I_p \cos(\omega t - \theta + 120^\circ)$$

$$\mathbf{I} = \mathbf{V} / \mathbf{Z}$$
$$\rightarrow \angle \mathbf{I} = \angle \mathbf{V} - \angle \mathbf{Z}$$

The total instantaneous power is

$$p = p_a + p_b + p_c$$

$$= v_{AN} i_a + v_{BN} i_b + v_{CN} i_c$$

$$= 3V_p I_p \cos \theta$$

$$\begin{aligned} &= 2V_p I_p [\cos \omega t \cos(\omega t - \theta) \\ &\quad + \cos(\omega t - 120^\circ) \cos(\omega t - \theta - 120^\circ) \\ &\quad + \cos(\omega t + 120^\circ) \cos(\omega t - \theta + 120^\circ)] \\ &= 2V_p I_p \left\{ \frac{\cos \theta + \cos(2\omega t - \theta)}{2} \right. \\ &\quad \left. + \frac{\cos \theta + \cos(2\omega t - \theta - 240^\circ)}{2} \right. \\ &\quad \left. + \frac{\cos \theta + \cos(2\omega t - \theta + 240^\circ)}{2} \right\} \\ &= 3V_p I_p \cos \theta \end{aligned}$$

$$p = p_a + p_b + p_c = 3V_p I_p \cos \theta$$

Thus, **the total instantaneous power** in a balanced three-phase system is **constant** – it does not change with time as the instantaneous power of each phase does.

This result is true whether the load is Y- or Δ -connected. This is one important reason for using a three-phase system to generate and distribute power.

The advantage over a single-phase AC system: **constant $p(t)$**

Since the total instantaneous power is independent of time,

$$Z_Y = Z \angle \theta \quad \theta \text{ is the power factor angle}$$

- (i) **Average power** per phase P_p for the Y- or Δ -connected load is $P/3$, or $P_p = V_P I_P \cos \theta$
- (ii) **Reactive power** per phase is $Q_p = V_P I_P \sin \theta$
- (iii) **Apparent power** per phase is $|S_P| = V_P I_P$
- (iv) **Complex power** per phase is $S_P = P_P + jQ_P = V_P I_P^*$
- (v) **Total average power** by AC power conservation is $P = P_a + P_b + P_c = 3P_P = 3V_P I_P \cos \theta$

The total average power:

For a Y-connected load (with Y-connected source),
 $I_L = I_P$ *but* $V_L = \sqrt{3}V_P$

For a Δ -connected load (with Δ -connected source),
 $I_L = \sqrt{3}I_P$ *but* $V_L = V_P$

Thus, $3V_P I_P \cos\theta = \sqrt{3}V_L I_L \cos\theta$

V_P : phase voltage
 I_P : phase current

V_L : line voltage
 I_L : line current

The total average power:

$$P = 3P_p = 3V_p I_p \cos \theta = \sqrt{3} V_L I_L \cos \theta$$

The total reactive power:

$$Q = 3Q_p = 3V_p I_p \sin \theta = \sqrt{3} V_L I_L \sin \theta$$

Recall:

$$Z_Y = Z \angle \theta; \quad \theta = \angle \mathbf{V}_p - \angle \mathbf{I}_p$$

θ is the power factor angle

The total complex power:

$$S = 3S_p = 3V_p \mathbf{I}_p^* = 3I_p^2 Z_p = \frac{3V_p^2}{Z_p^*}$$

where Z_p is the load impedance per phase.

$$S = P + jQ = 3V_p I_p \cos \theta + j3V_p I_p \sin \theta = \sqrt{3} V_L I_L \cos \theta + j\sqrt{3} V_L I_L \sin \theta$$

V_p , I_p , V_L , and I_L are **all rms values** and that θ is the angle of the load impedance or the angle between the phase voltage and the phase current.

The total average power:

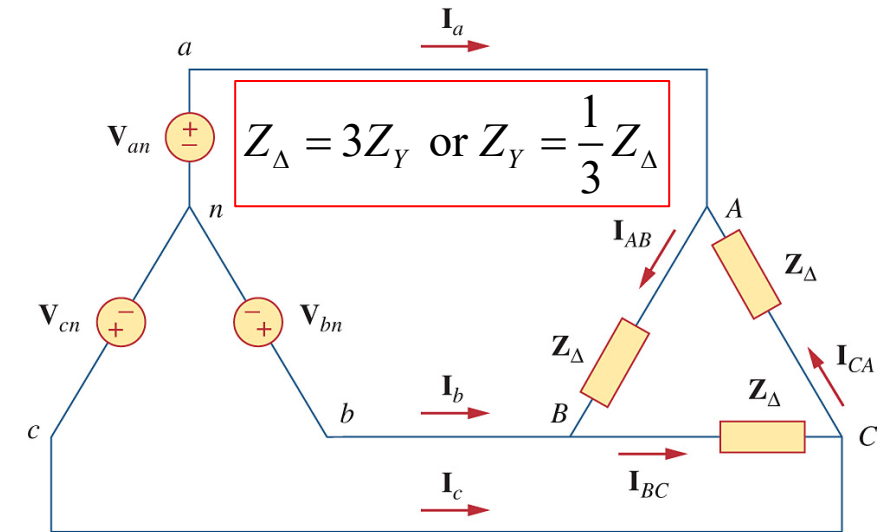
For a Y-Y connection, $I_L = I_P$ *but* $V_L = \sqrt{3}V_P$

For a Δ - Δ connection, $I_L = \sqrt{3}I_P$ *but* $V_L = V_P$

$$P = 3P_P = 3V_P I_P \cos\theta = \sqrt{3}V_L I_L \cos\theta$$

How about other connections?

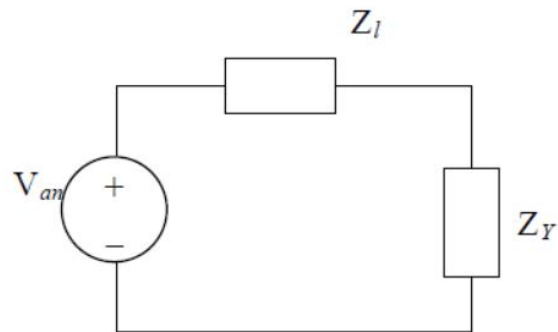
12.29 A balanced three-phase Y- Δ system has $\mathbf{V}_{an} = 240\angle 0^\circ$ V rms and $\mathbf{Z}_{\Delta} = 51 + j45 \, \Omega$. If the line impedance per phase is $0.4 + j1.2 \, \Omega$, find the total complex power delivered to the load.



- 12.29** A balanced three-phase Y- Δ system has $\mathbf{V}_{an} = 240\angle 0^\circ$ V rms and $\mathbf{Z}_{\Delta} = 51 + j45 \, \Omega$. If the line impedance per phase is $0.4 + j1.2 \, \Omega$, find the total complex power delivered to the load.



Replace the delta load with a Y load, $\mathbf{Z}_Y = \mathbf{Z}_{\Delta}/3$

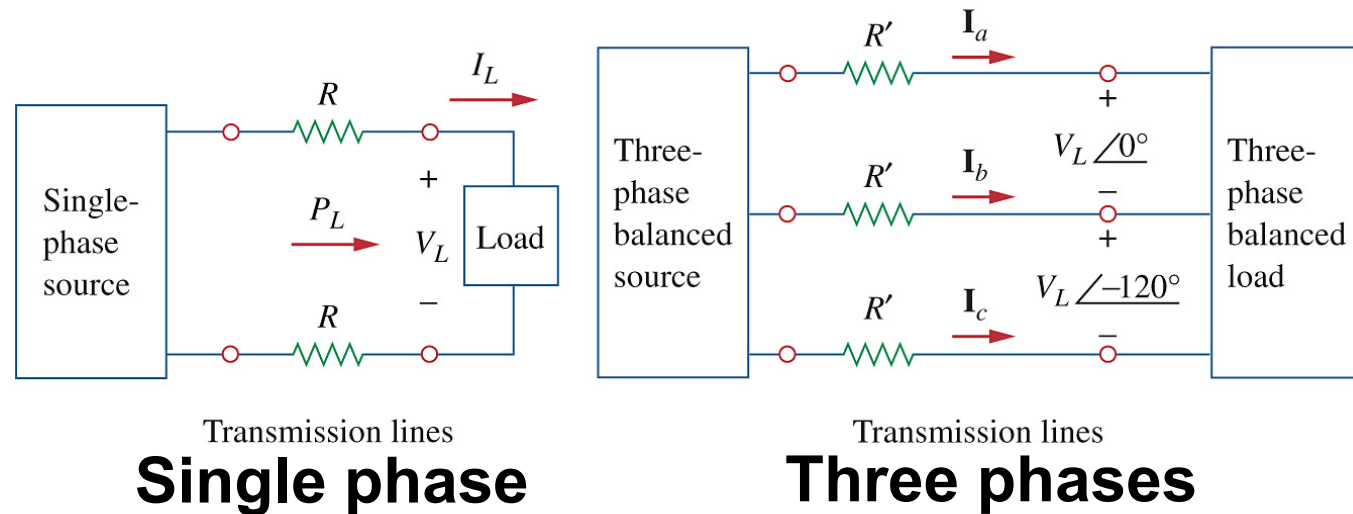


$$\mathbf{I}_a = \mathbf{V}_{an} / [\mathbf{Z}_Y + \mathbf{Z}_l] = 240 / [17 + j15 + 0.4 + j1.2] = 240 / [17.4 + j16.2] = 240 / 23.77 = 10.095$$

$$\begin{aligned} \mathbf{S} &= 3[(\mathbf{I}_a)^2(17 + j15)] = 3 \times 101.91(17 + j15) \\ &= [5.197 + j4.586] \text{ kVA.} \end{aligned}$$

Three phase system - Advantage 2:

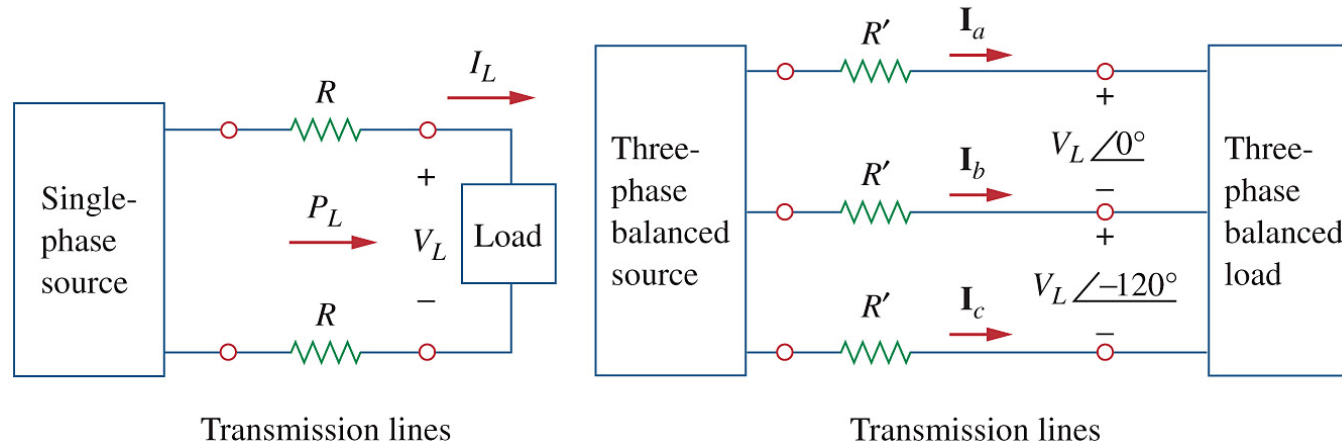
The three-phase system uses a lesser amount of wire than the single-phase system for the same line voltage V_L and **the same absorbed power P_L** → Economical



1. The same total power delivered P_L
2. The same line voltage V_L

Recall: Average power in the rms form in Chapter 11

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$



Assume that the load is purely resistive

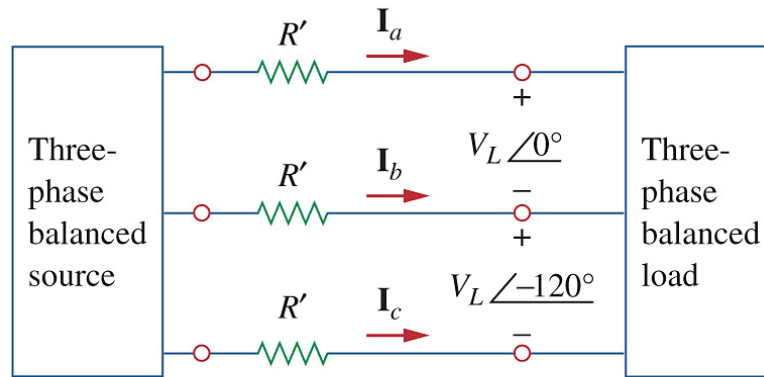
For the single-phase **two-wire system**, $P_L = V_L I_L$, the power loss in the two wires is

$$P_{\text{loss}} = 2I_L^2 R = 2 \left(\frac{P_L}{V_L} \right)^2 R$$

2 wires

Recall: Average power in the rms form in Chapter 11

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$



Transmission lines

I'_L : line current
 R' : line resistance

$$I'_L = |\mathbf{I}_a| = |\mathbf{I}_b| = |\mathbf{I}_c|$$

$$= P_L / \sqrt{3} V_L$$

For the three-phase wire system, $P_L = \sqrt{3} V_L I'_L$, the power loss in the three wires is

$$P'_{\text{loss}} = 3 I'^2_L R' = 3 \left(\frac{P_L}{\sqrt{3} V_L} \right)^2 R' = \left(\frac{P_L}{V_L} \right)^2 R' \quad \boxed{\text{3 wires}}$$

The equations show that for **the same total power delivered P_L** and **the same line voltage V_L** ,

$$P_{loss} = 2I_L^2 R = 2 \left(\frac{P_L}{V_L} \right)^2 R \quad \boxed{\text{2 wires}}$$

$$P'_{loss} = 3I_L'^2 R' = 3 \left(\frac{P_L}{\sqrt{3}V_L} \right)^2 R' = \left(\frac{P_L}{V_L} \right)^2 R' \quad \boxed{\text{3 wires}}$$

$$\frac{P_{loss}}{P'_{loss}} = \frac{2R}{R'} = \frac{2\rho l / (\pi r^2)}{\rho l / (\pi r'^2)} = \frac{2r'^2}{r^2}$$

$$R = \frac{\rho L}{A} \quad \begin{array}{l} \rho = \text{resistivity} \\ L = \text{length} \\ A = \text{cross sectional area} \end{array}$$

where ρ is resistivity; l is length of transmission line; and r is radius of the transmission line

If we consider **the same power loss** in both systems,

$$\frac{P_{loss}}{P'_{loss}} = \frac{2r'^2}{r^2} = 1 \quad \text{thus, } r^2 = 2r'^2$$

Two wires

Three wires

$$\frac{\text{Material for single-phase}}{\text{Material for three-phase}} = \frac{2(\pi r^2 l)}{3(\pi r'^2 l)} = \frac{2r^2}{3r'^2} = \frac{4}{3}$$

Two wires

Material volume needed



<https://www.energy.gov/sites/prod/files/2015/12/f28/united-states-electricity-industry-primer.pdf>

The second major advantage of three-phase systems for power distribution: **Economical**

Summary of the advantages of 3- ϕ system

1. The total instantaneous power in a balanced three-phase system is constant.
2. The material (transmission line) to deliver the same power and to tolerate the same loss needed is $\frac{3}{4}$ times less.

Example 12.8

$$\begin{aligned} S &= P + jQ = 3V_P I_P \cos \theta + j3V_P I_P \sin \theta \\ &= \sqrt{3}V_L I_L \cos \theta + j\sqrt{3}V_L I_L \sin \theta \end{aligned}$$

Two balanced loads are connected to a 240-kV rms 60-Hz line, as shown in Fig. 12.22(a). Load 1 draws 30 kW at a power factor of 0.6 lagging, while load 2 draws 45 kVAR at a power factor of 0.8 lagging. Assuming the *abc* sequence, determine: (a) the complex, real, and reactive powers absorbed by the combined load, (b) the line currents, and (c) the kVAR rating of the three capacitors Δ -connected in parallel with the load that will raise the power factor to 0.9 lagging and the capacitance of each capacitor.

Line voltage

30kW
pf=0.6 lagging

45kVAR
pf=0.8 lagging

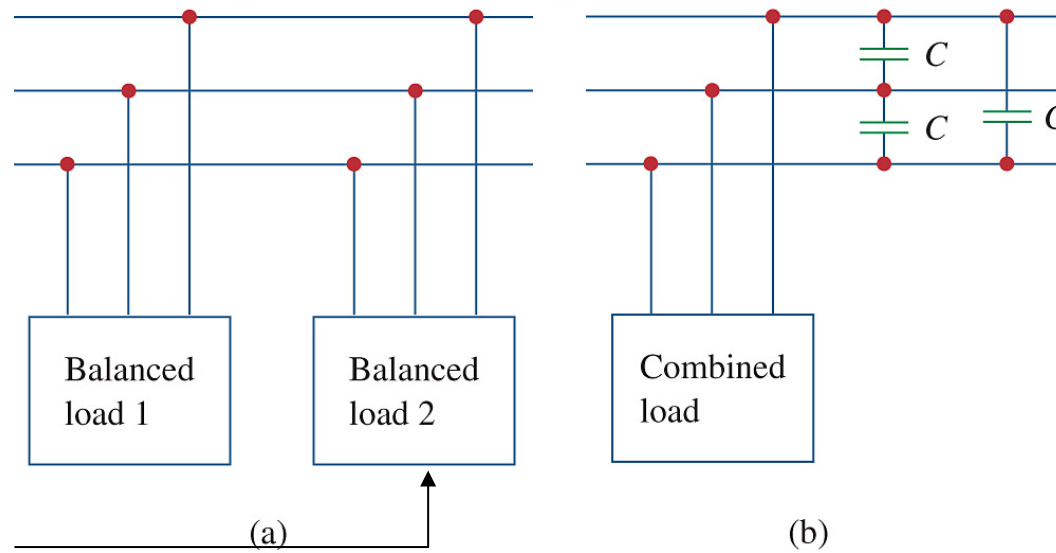


Figure 12.22 (a) The original balanced loads, (b) the combined load with improved power factor.

Load 1:
30kW
pf=0.6 lagging

Load 2:
45kVAR
pf=0.8 lagging

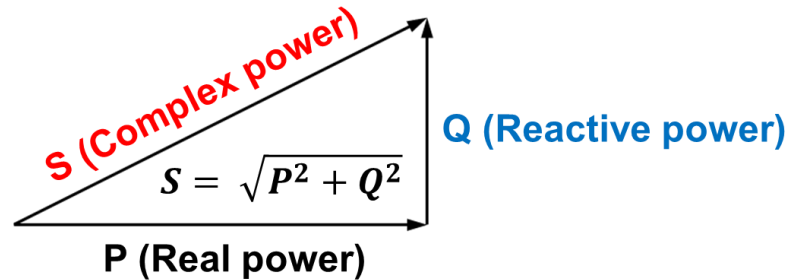
$$V_L = 240\text{kV}, 60\text{ Hz}$$

$$S = P + jQ = 3V_P I_P \cos \theta + j3V_P I_P \sin \theta$$

$$= \sqrt{3}V_L I_L \cos \theta + j\sqrt{3}V_L I_L \sin \theta$$

(a) S, P, Q

Load 1: 30kW, pf=0.6 lagging



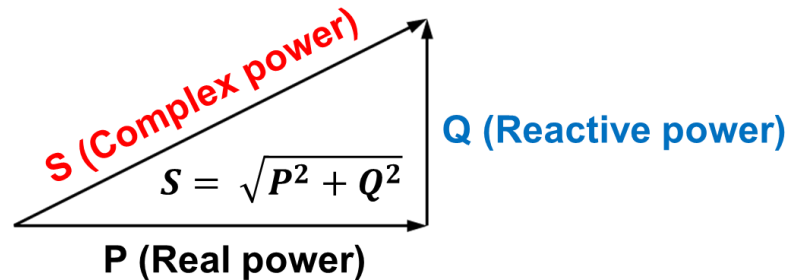
$$\cos \theta_1 = 0.6 \rightarrow \theta_1 = 53.13^\circ$$

$$\tan 53.13^\circ = Q/P$$

Thus, $Q = 40\text{k [VAR]}$

$$\mathbf{S_1 = 30k + j40k [VA]}$$

Load 2: 45kVAR, pf=0.8 lagging



$$\cos \theta_2 = 0.8 \rightarrow \theta_2 = 36.87^\circ$$

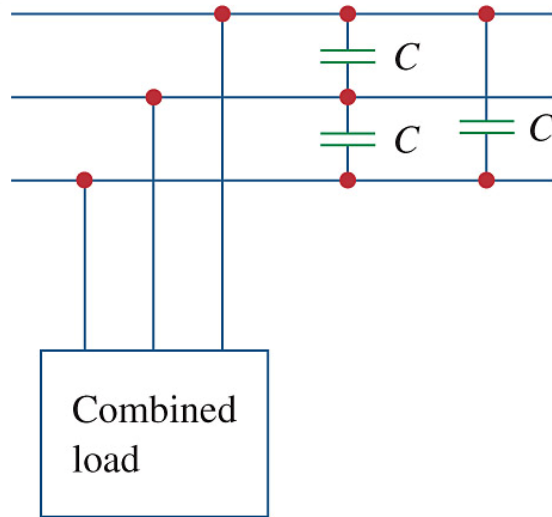
$$\tan 36.87^\circ = Q/P$$

Thus, $P = 60\text{k [W]}$

$$\mathbf{S_1 = 60k + j45k [VA]}$$

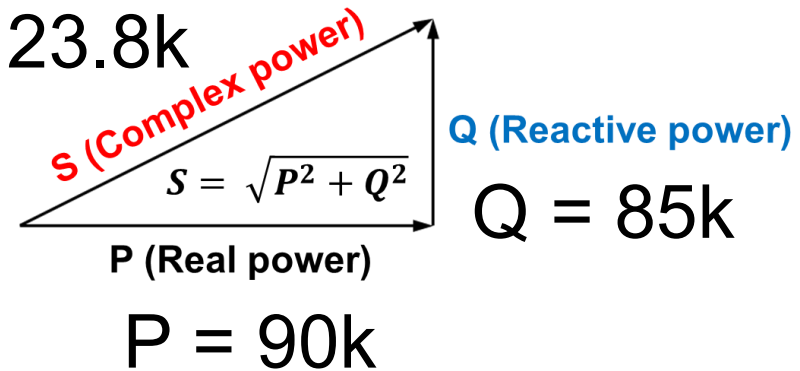
Complex power absorbed by the combined load, therefore, is $\mathbf{S = 90k + j85k [VA]}$ where $P = 90\text{kW}$ and $Q = 85\text{kVA}$

(b) Line currents



240 kV rms 60-Hz line $\rightarrow V_L = 240$ kV
 S from (a) is $90k + j85k$

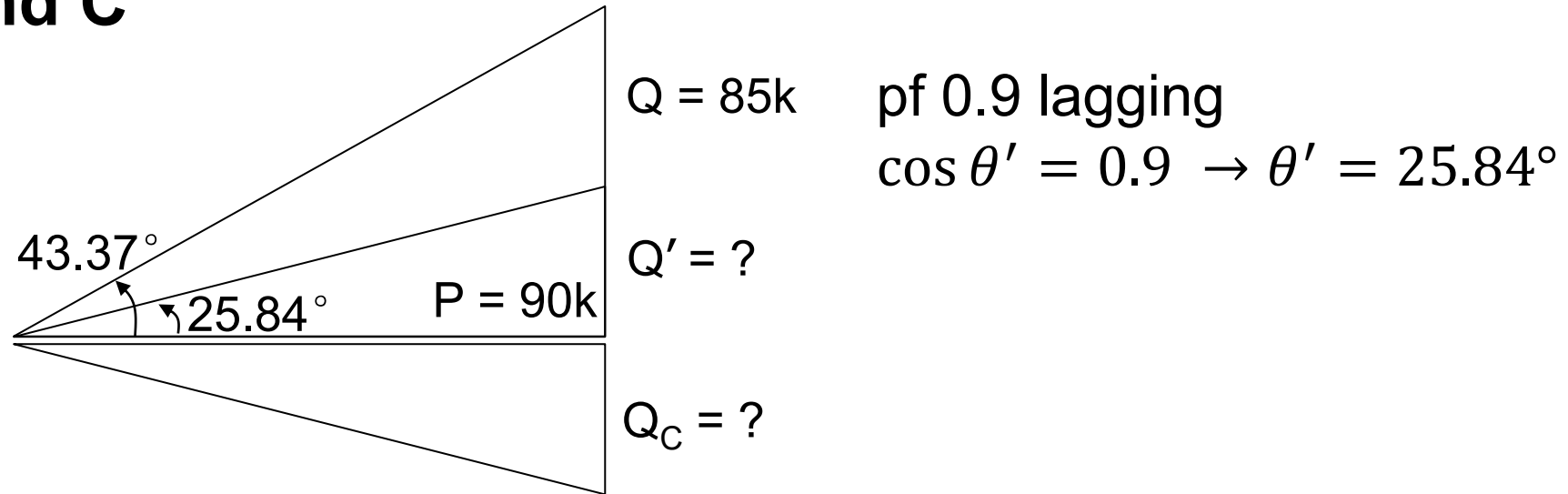
$$|S| = 123.8k$$



$$P = \sqrt{3} V_L I_L \cos \theta \rightarrow 90k = \sqrt{3} 240k I_L \cos 90^\circ / 123.8$$

Thus, $I_L = 0.296$ A or 0.296 mA

(c) Q_C and C



$$Q' \rightarrow \tan 25.84 = \frac{Q'}{P(90k)} \rightarrow Q' = 43.56k$$

$$Q_C = Q - Q' = 85k - 43.56k = \mathbf{41.44k \text{ VAR}}$$

→ Each capacitor has an effect of 13.81k VAR

$Q_C = I^2 X = \omega C V^2$ for purely capacitive case, and thus,

$$C = \frac{Q_C}{\omega(=2\pi f)V^2} = \mathbf{6.36 \times 10^{-10} [F]}$$